Curs 9

Filtre

Disclaimer

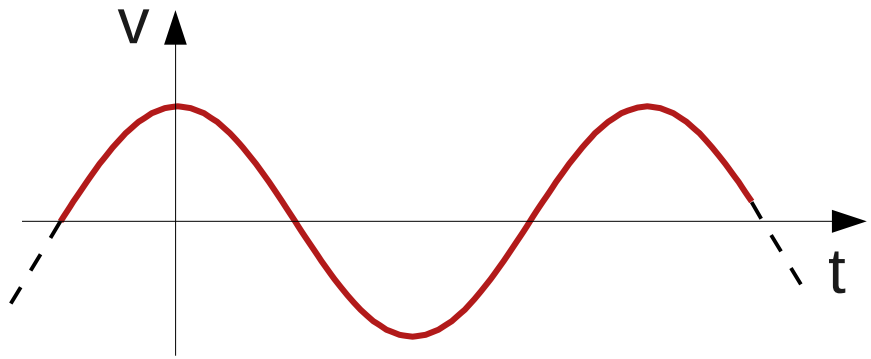
- *Slide-urile acestea reprezintă un rezumat.*
- *Slide-urile nu acoperă toate explicațiile, simulările, demonstrațiile oferite în timpul cursului.*
- *Slide-urile nu limitează în vreun fel materia cerută la examen.*
- *Pentru varianta completă, sunteți așteptați la curs*

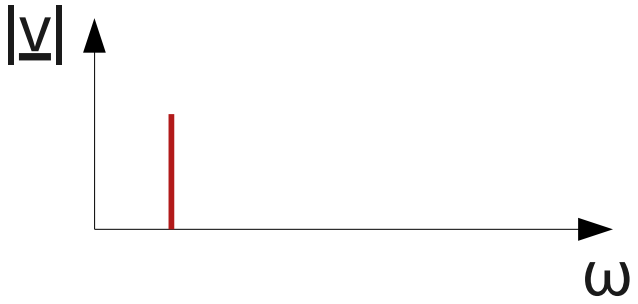
Joi 10:00 / 12:00 – on-line, Microsoft teams

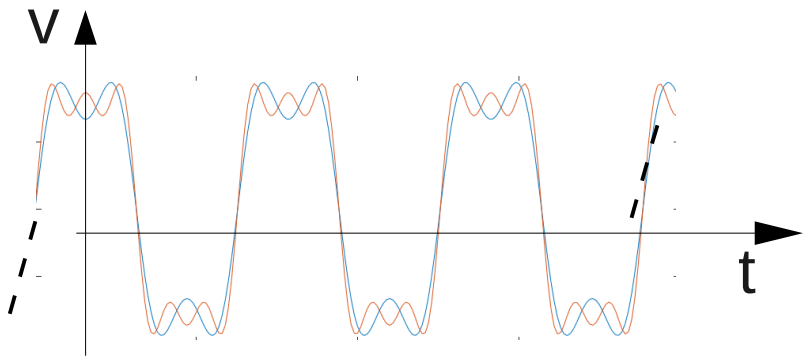
Luni 12:00/14:00, on-line, Microsoft teams

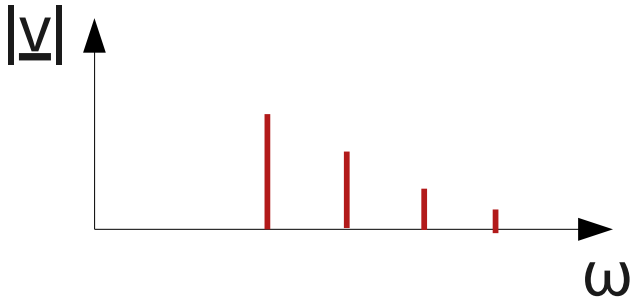
Intro

Intro - semnale

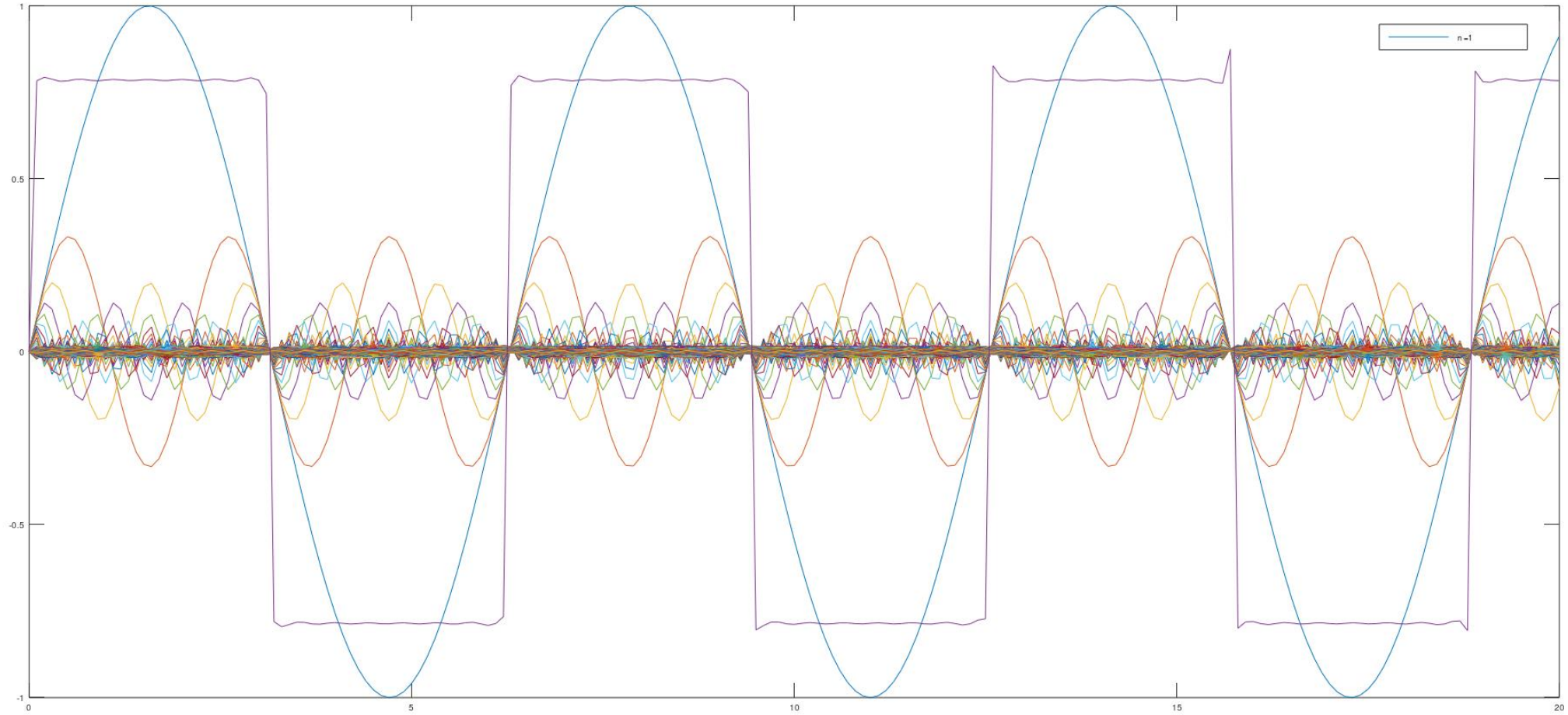




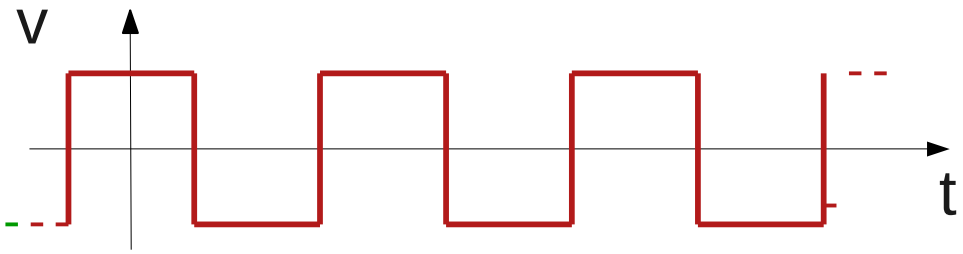


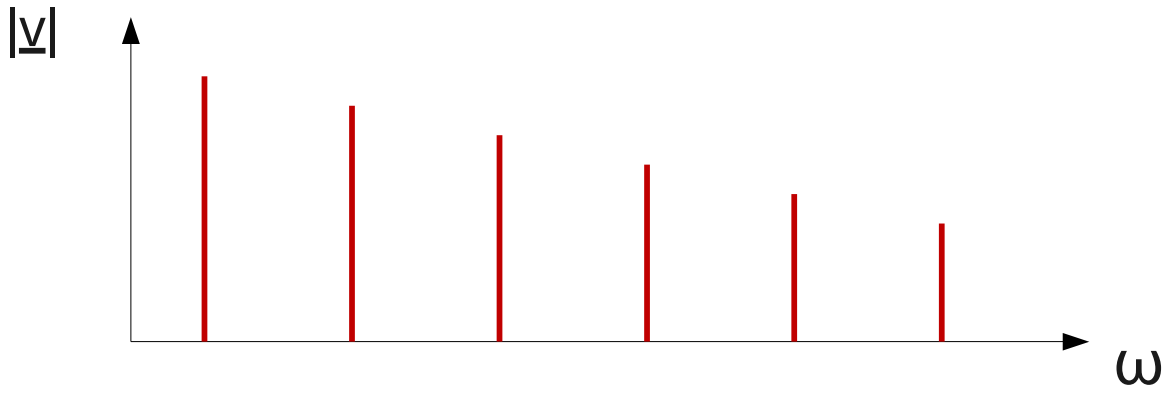


Intro - semnale

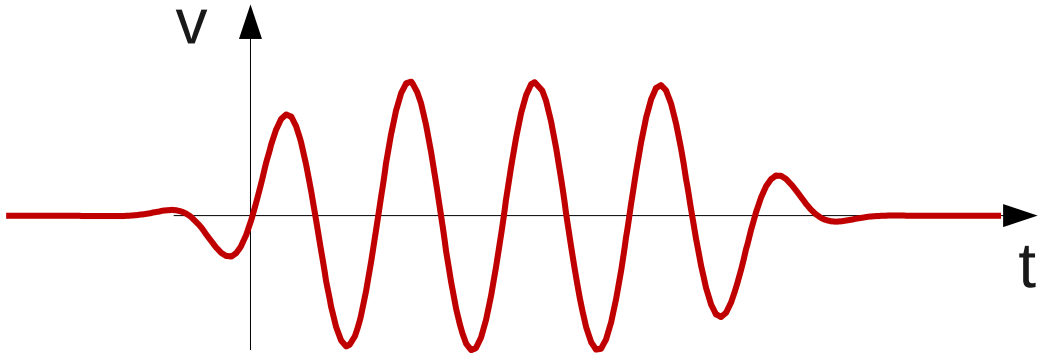


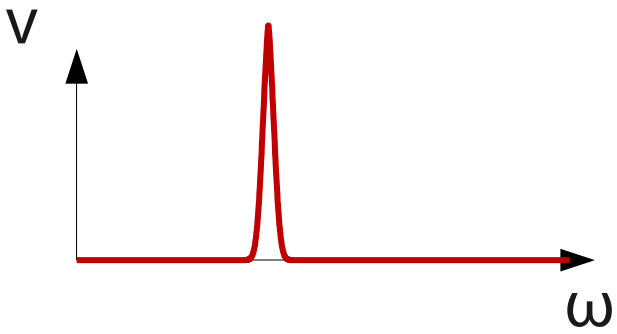
Intro - semnale

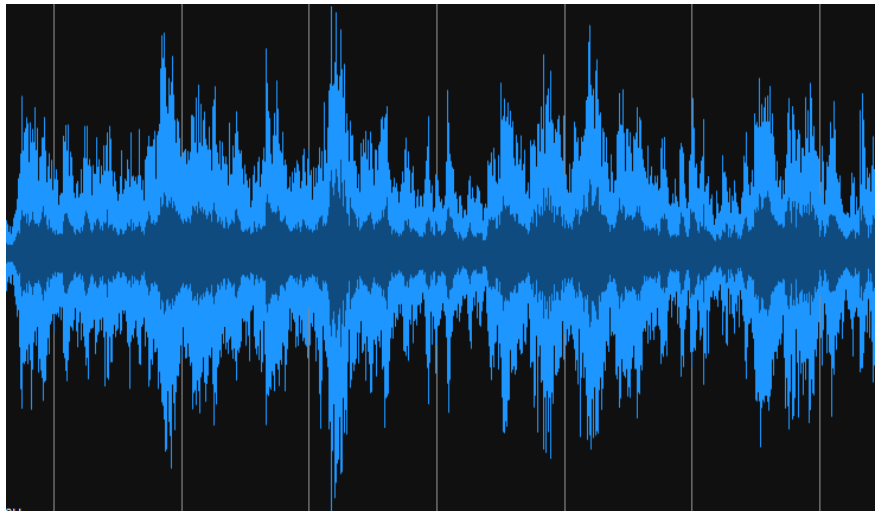


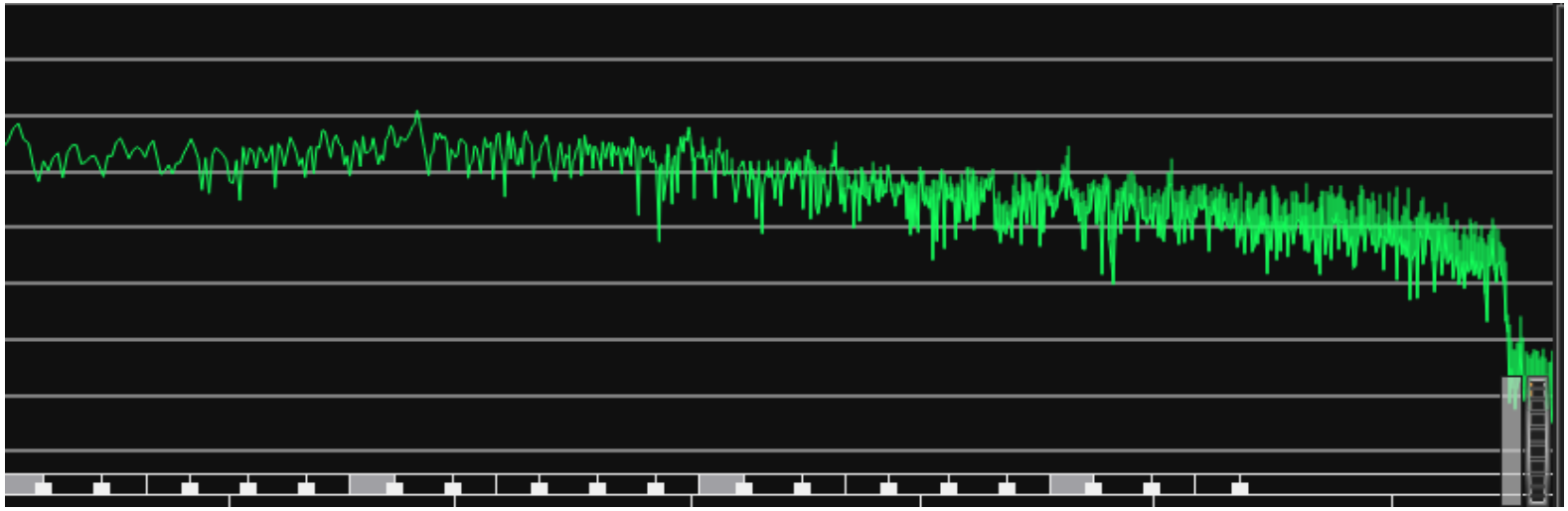


Intro - semnale



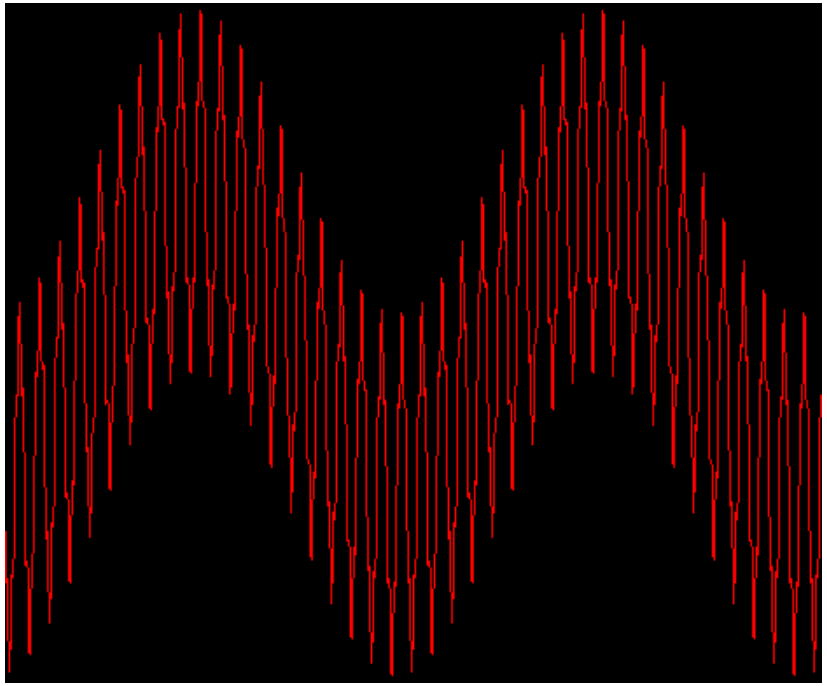


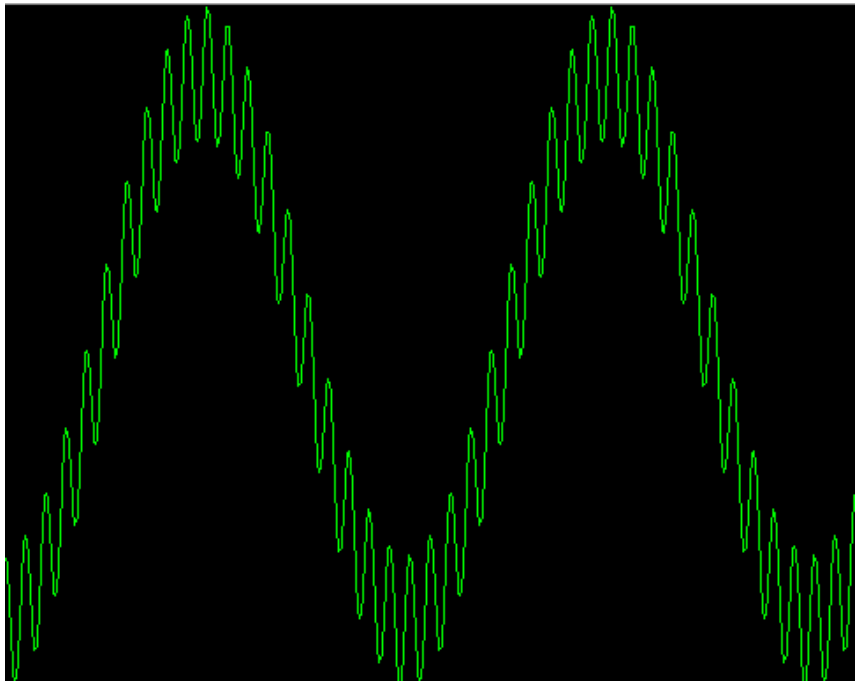


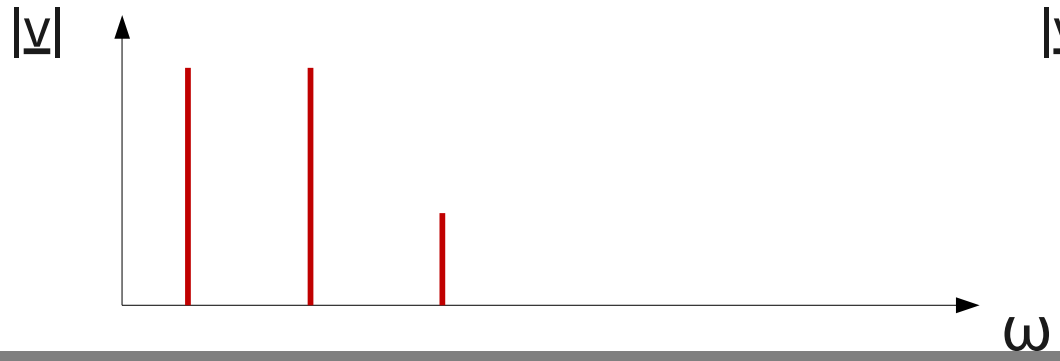


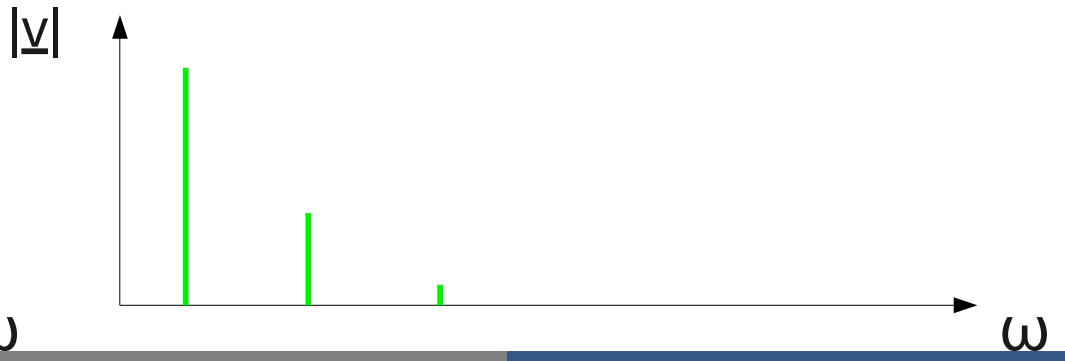
Filtre - introducere

Filtru trece jos









Generalități

- Un filtru electronic este un circuit electronic ce modifică amplitudinea și caracteristica de fază a unui semnal, în funcție de frecvența acestuia.
- În mod ideal, un filtru modifică amplitudinea anumitor componente, în funcție de frecvența lor
- Utilizări:
  - eliminarea componentelor de frecvență nedorită (ex: zgomote)
  - evidențierea componentelor de frecvență de interes
- Întrucât comportamentul unui filtru depinde de frecvența semnalului, se utilizează analiza în domendiu frecvență

$$H(s) = \frac{V_{OUT}(s)}{V_{IN}(s)}$$

Exemplu

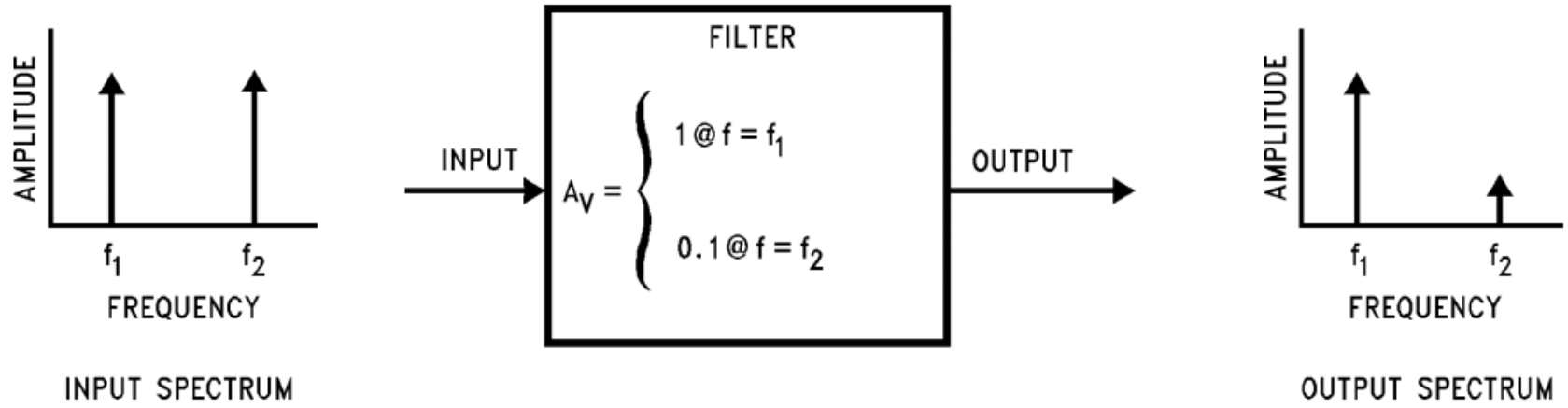


FIGURE 1. Using a Filter to Reduce the Effect of an Undesired Signal at Frequency f_2, while Retaining Desired Signal at Frequency f_1

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Răspunsul în frecvență și răspunsul în fază

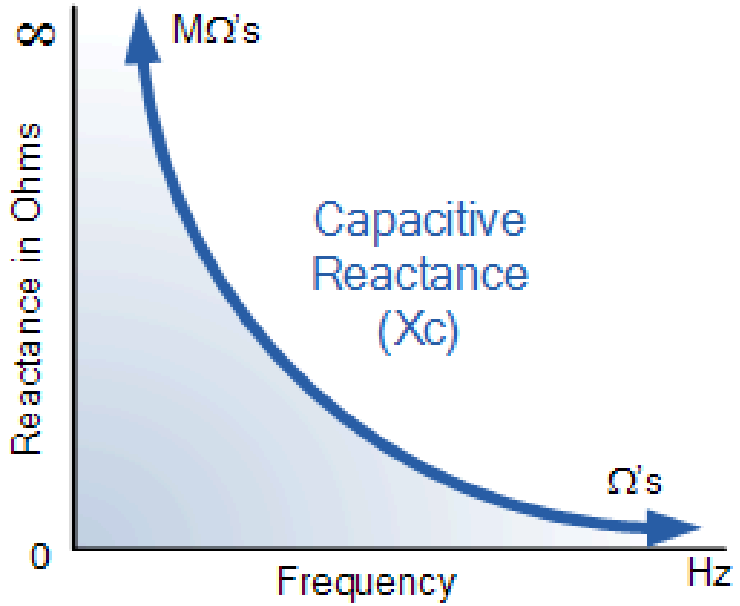
$$|H(j\omega)| = \left|\frac{V_{OUT}(j\omega)}{V_{IN}(j\omega)}\right|$$

$$\arg H(j\omega) = \arg \frac{V_{OUT}(j\omega)}{V_{IN}(j\omega)}$$

© A Basic Introduction to Filters - Active, Passive, and Switched-Capacitor

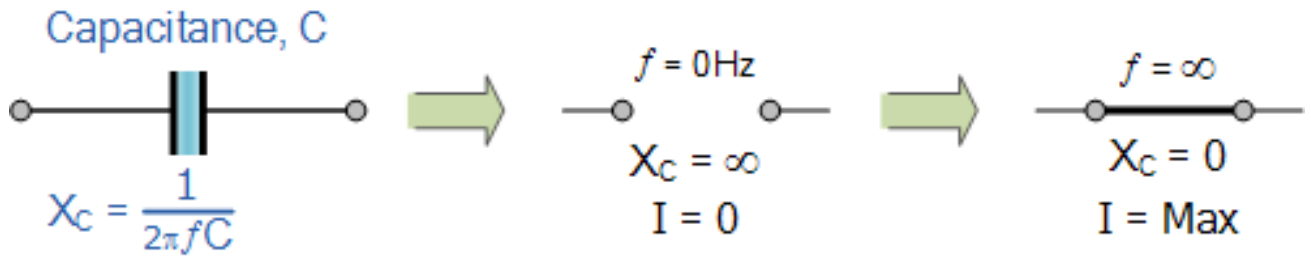
National Semiconductor Application Note 779 Kerry Lacanette April 1991

Recapitulare noțiuni necesare



$$Xc = \frac{1}{2\pi fC}$$

Xc = Capacitive Reactance in Ohms, (Ω)
π (pi) = 3.142 or 22/7
f = Frequency in Hertz, (Hz)
C = Capacitance in Farads, (F)



© http://www.electronics-tutorials.ws/filter/filter_1.html

Recapitulare noțiuni necesare

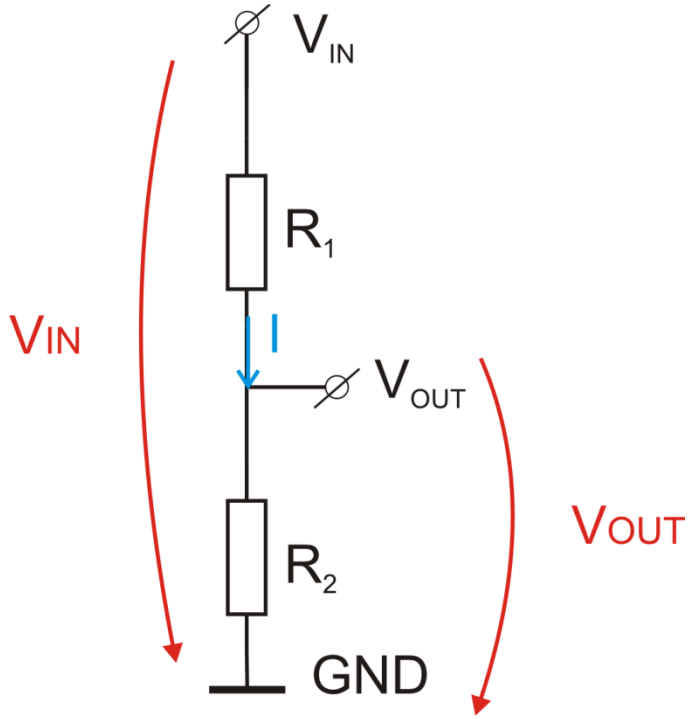
- $V_{out} = V_{in} (R_2 / (R_1 + R_2))$
- *Atenție! Formula se poate aplica doar dacă curentul din ramura cu V_{out} este mult mai mic decât cel prin R_1 și R_2!*

Demonstrație:

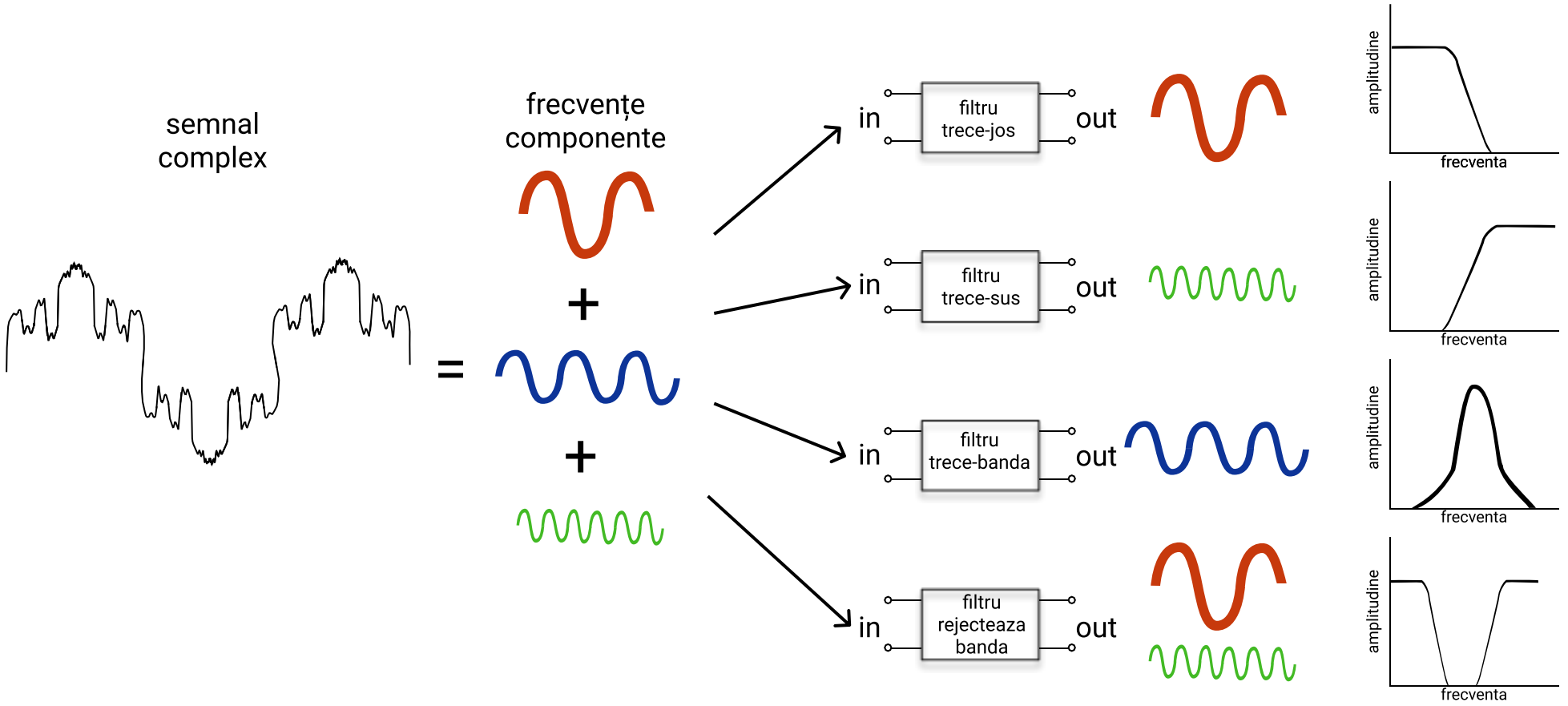
$V_{in} = I * R_1 + I * R_2 = I * (R_1 + R_2)$ *(1)*

$V_{out} = I * R_2$ *(2)*

Din *(2) / (1)* => $V_{out} = V_{in} (R_2 / (R_1 + R_2))$



Tipuri de filtre si aplicatii


semnal
complex
=
frecvențe
componente
+
+
in
filtru
trece-jos
out
in
filtru
trece-sus
out
in
filtru
trece-banda
out
in
filtru
rejecteaza
banda
out
amplitudine
frecventa
amplitudine
frecventa
amplitudine
frecventa
amplitudine
frecventa

Exemple aplicatii

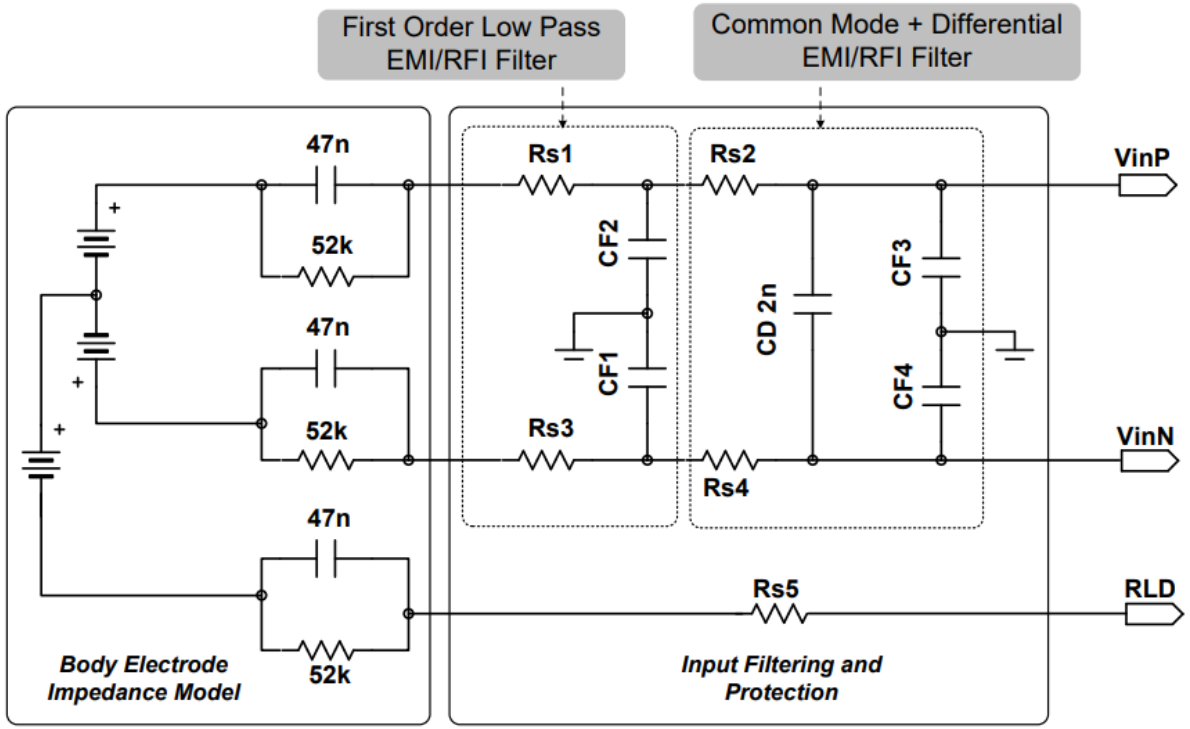
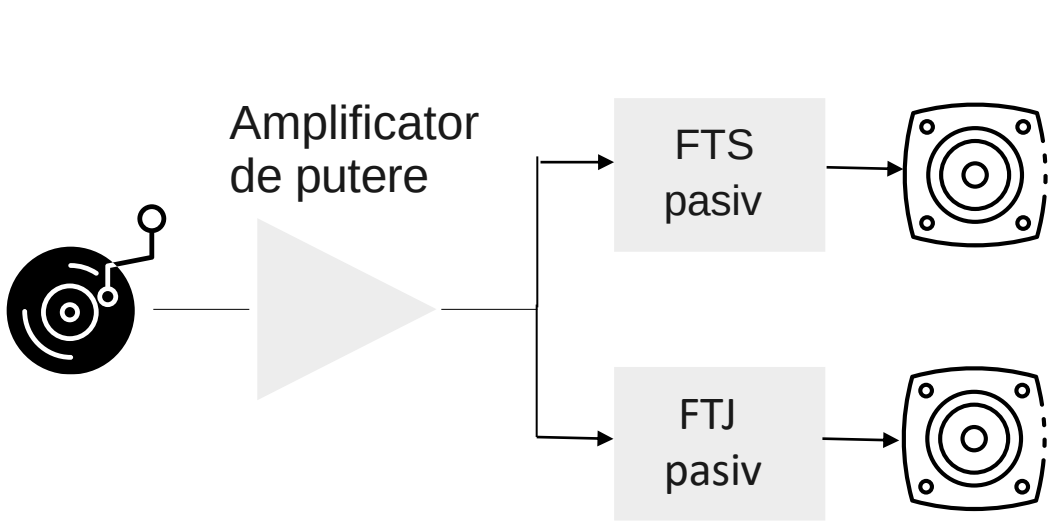
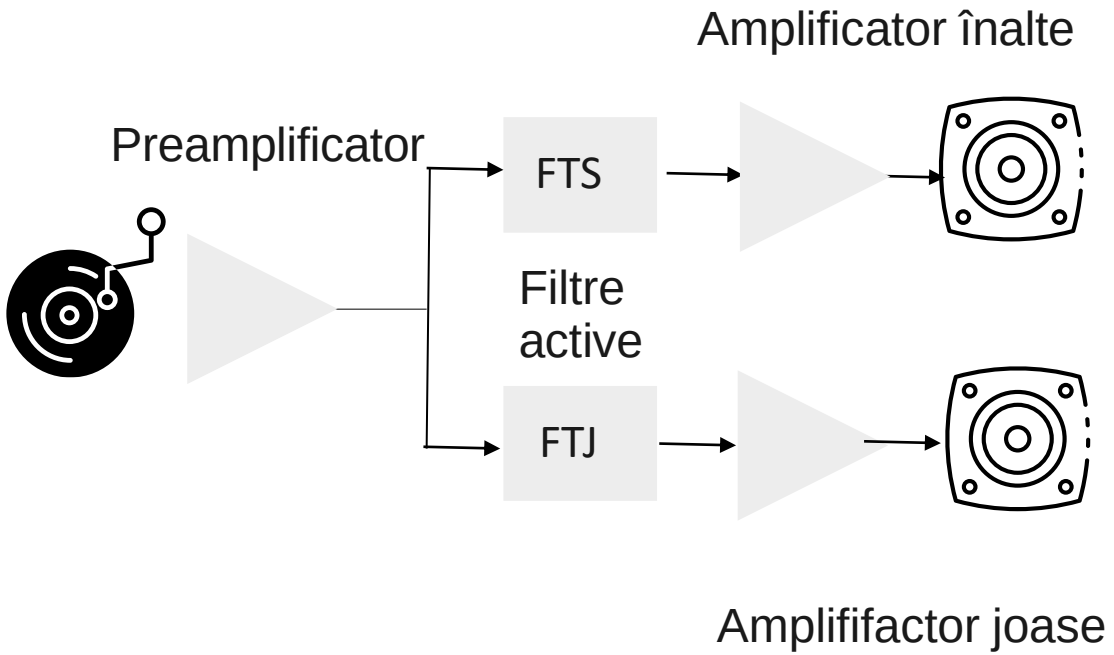


Figure 6: Input Protection and Filtering for ECG Front End

© TI Precision Designs: Verified Design Ultra Low Power, 18bit Precision ECG Data Acquisition System

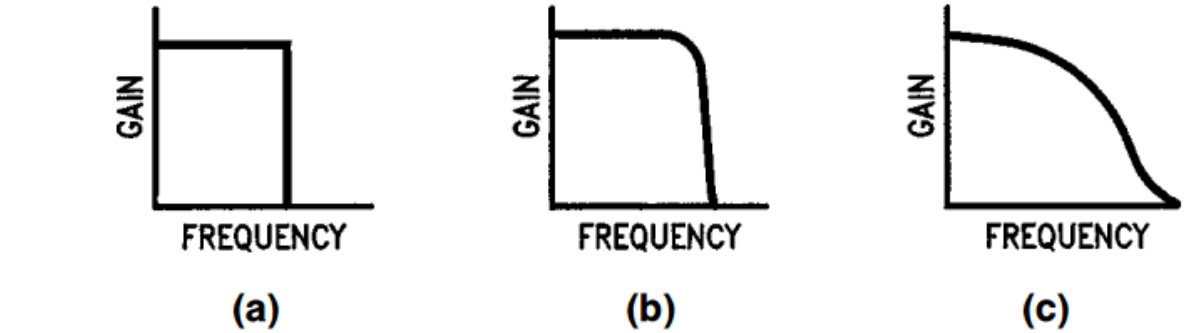
Exemple aplicatii



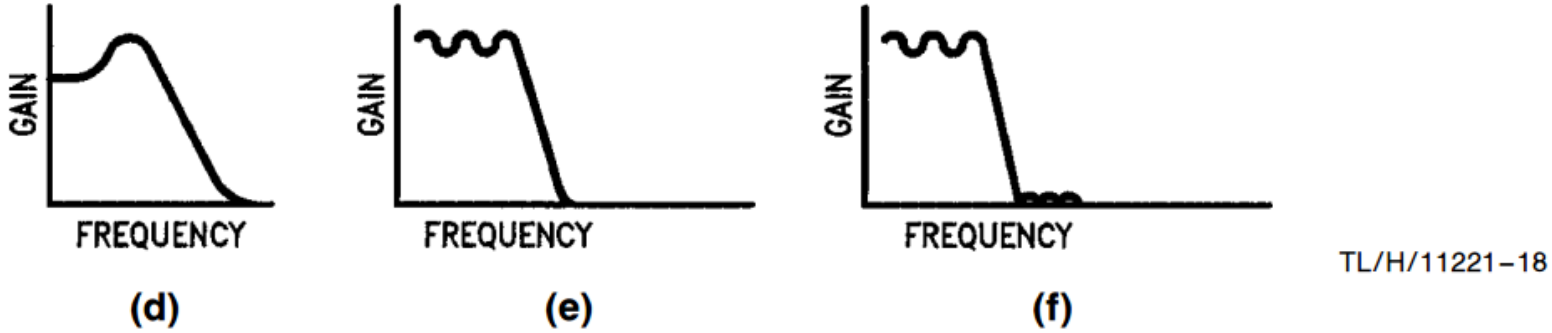


Filtru trece jos

Filtru trece jos



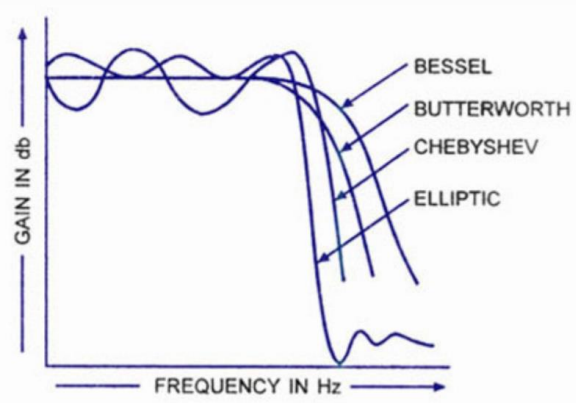
TL/H/11221–17



TL/H/11221–18

FIGURE 11. Examples of Low-Pass Filter Amplitude Response Curves

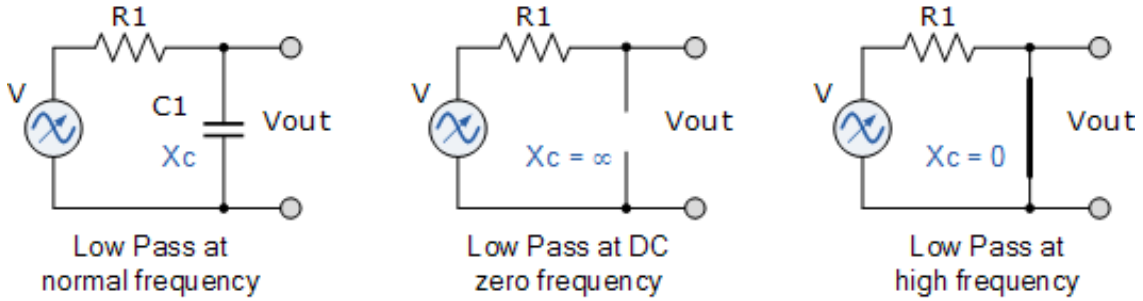
Filtru trece jos



Frequency Response Curve

© http://www.idc-online.com/technical_references/pdfs/electronic_engineering/Types_of_Active_Filters.pdf

FTJ (LPF) pasiv de ordin 1



$$V_{out} = V_{in} \frac{X_c}{\sqrt{R^2 + {X_c}^2}}$$

unde $X_c = \frac{1}{2\pi f C}$

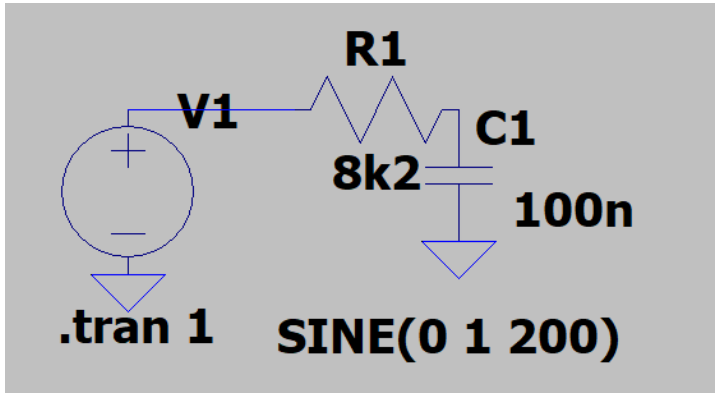
FTJ pasiv, ord 1, 200Hz

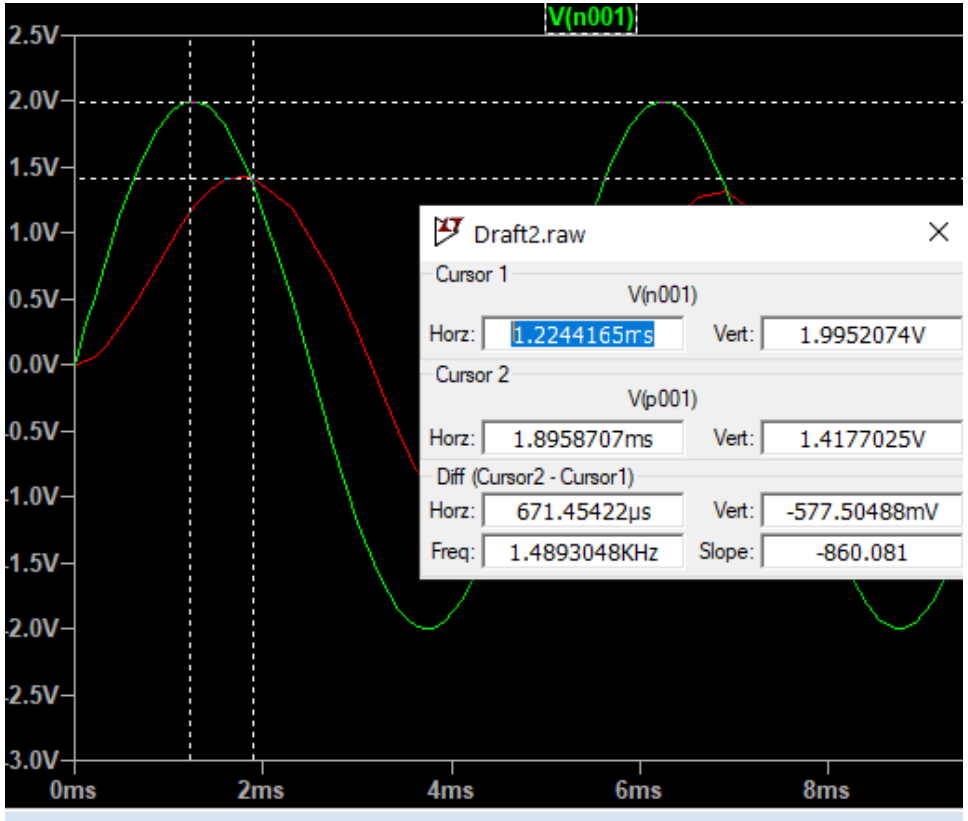
- R = 8k2Ω
- C = 100nF
- Vin = 2V
- Pentru f = 200Hz
  - Xc = 1 / (2 * 3.14 * 200 * 100 * 10^-9) = 7962
  - $V_{out} = 2V\ x\ \frac{7962}{\sqrt{8200^2 + 7962^2}} = 1{,}39V \cong 2V\ x\ \frac{1}{\sqrt{2}} = 1{,}41V$

http://sim.okawa-denshi.jp/en/CRtool.php

obs:

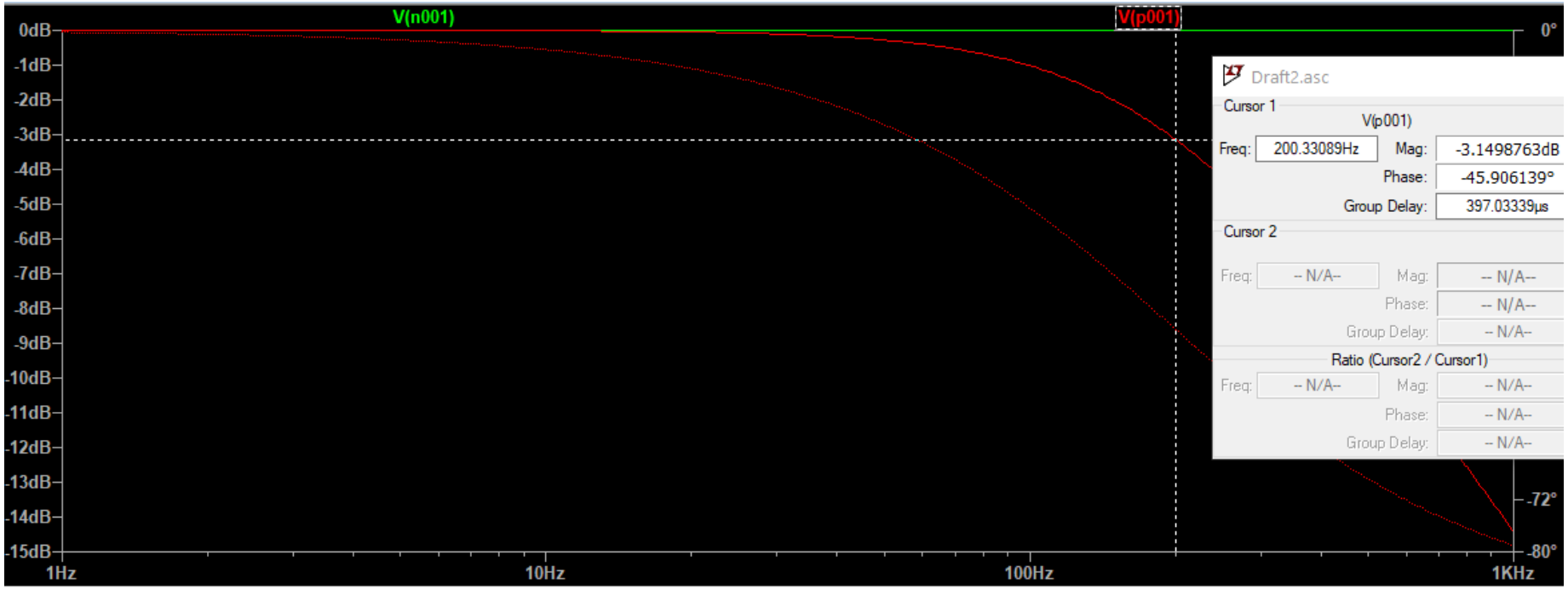
$$20 \lg\left(\frac{1}{\sqrt{2}}\right) \cong -3dB$$



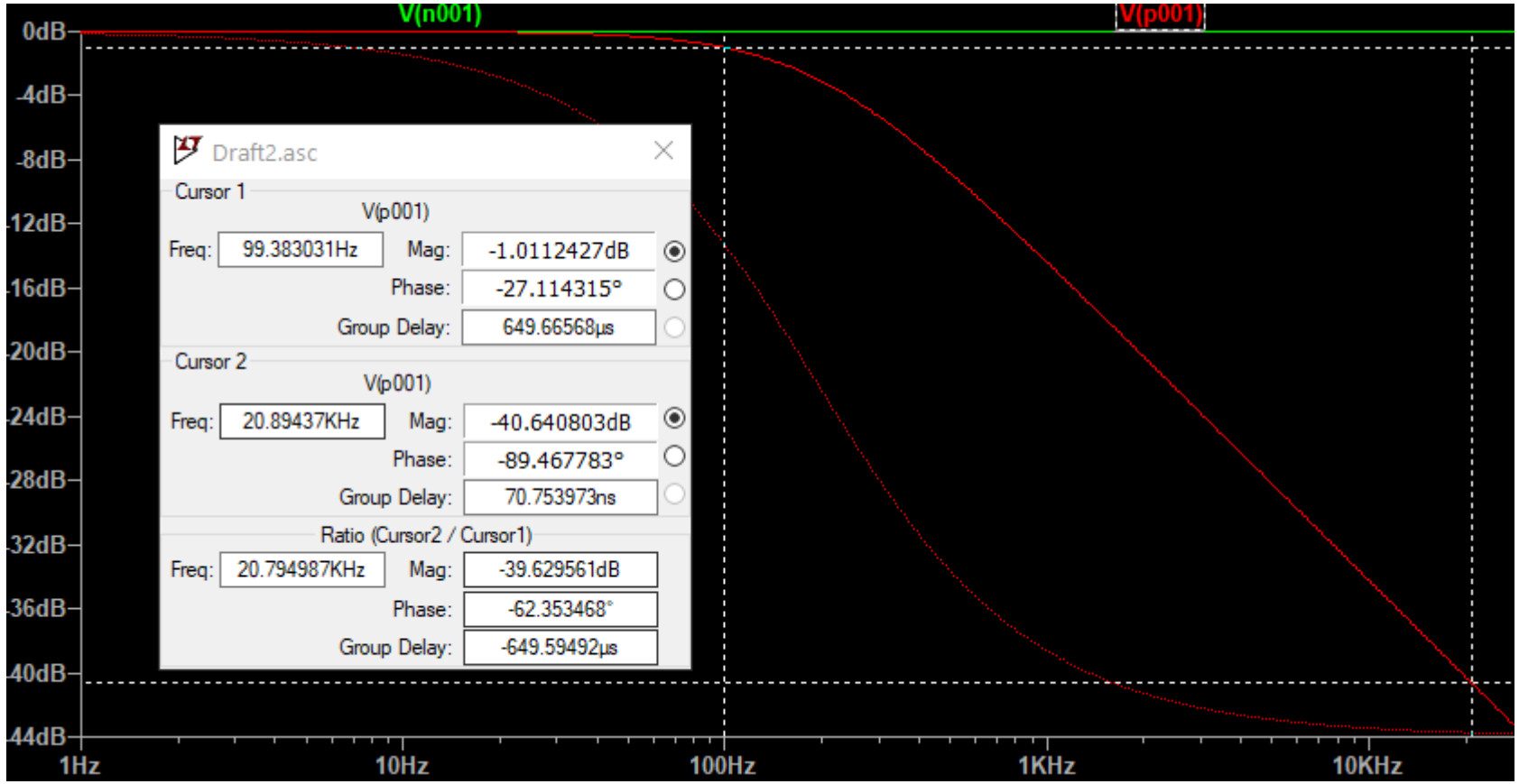


Banda de trecere, punctul de -3 dB

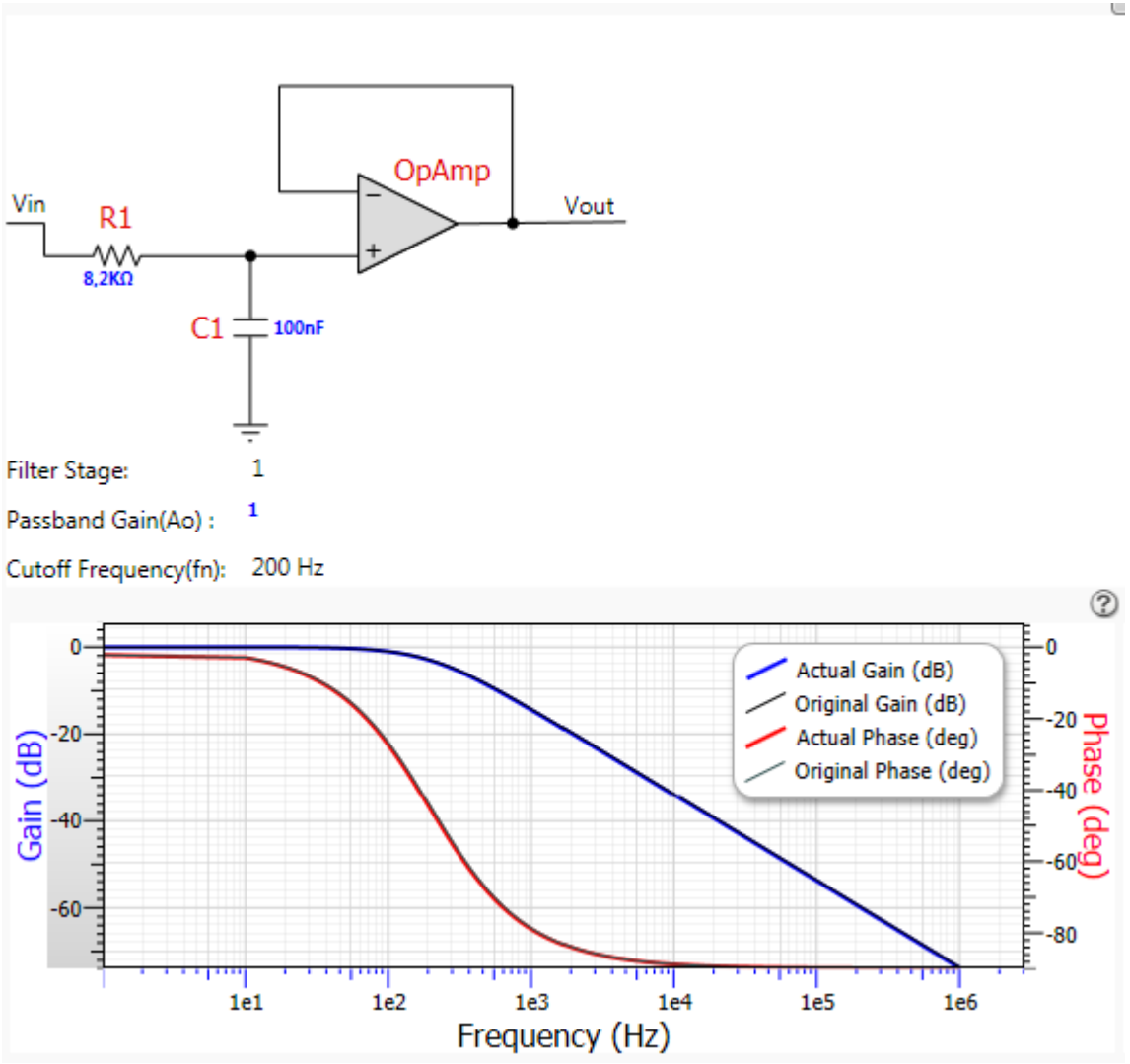
- Banda de trecere – punctul unde amplificarea de putere scade la jumătate
- Tensiunea scade de radical din 2 ori
- Se mai numeste punctul de -3 dB



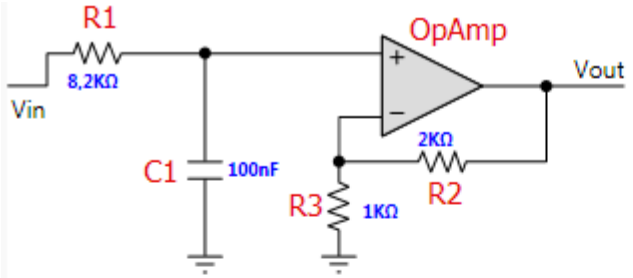
Pentru un filtru de ordin I: panta –20 dB / decadă



FTJ ord 1, cu buffer, 200Hz

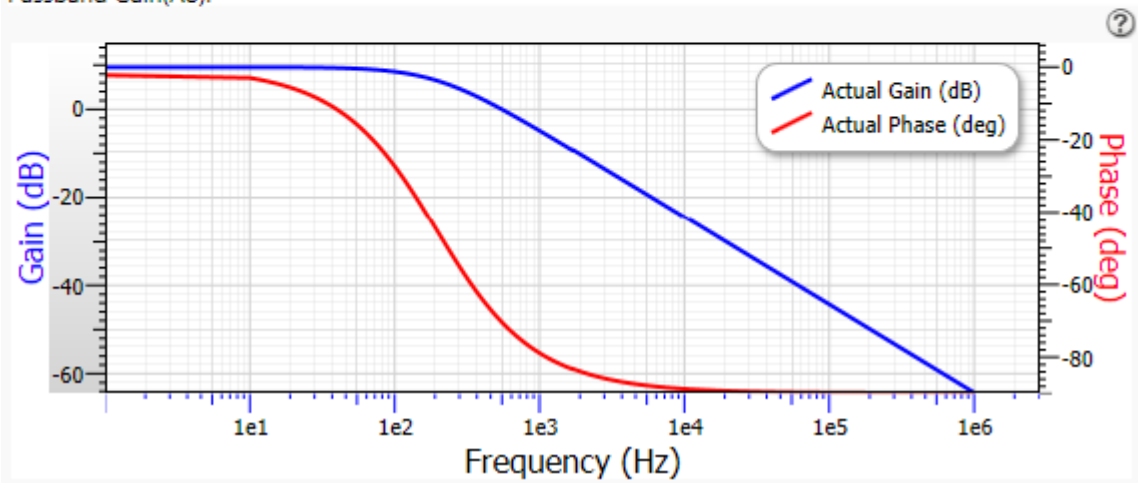


FTJ ord 1, gain 3, 200Hz

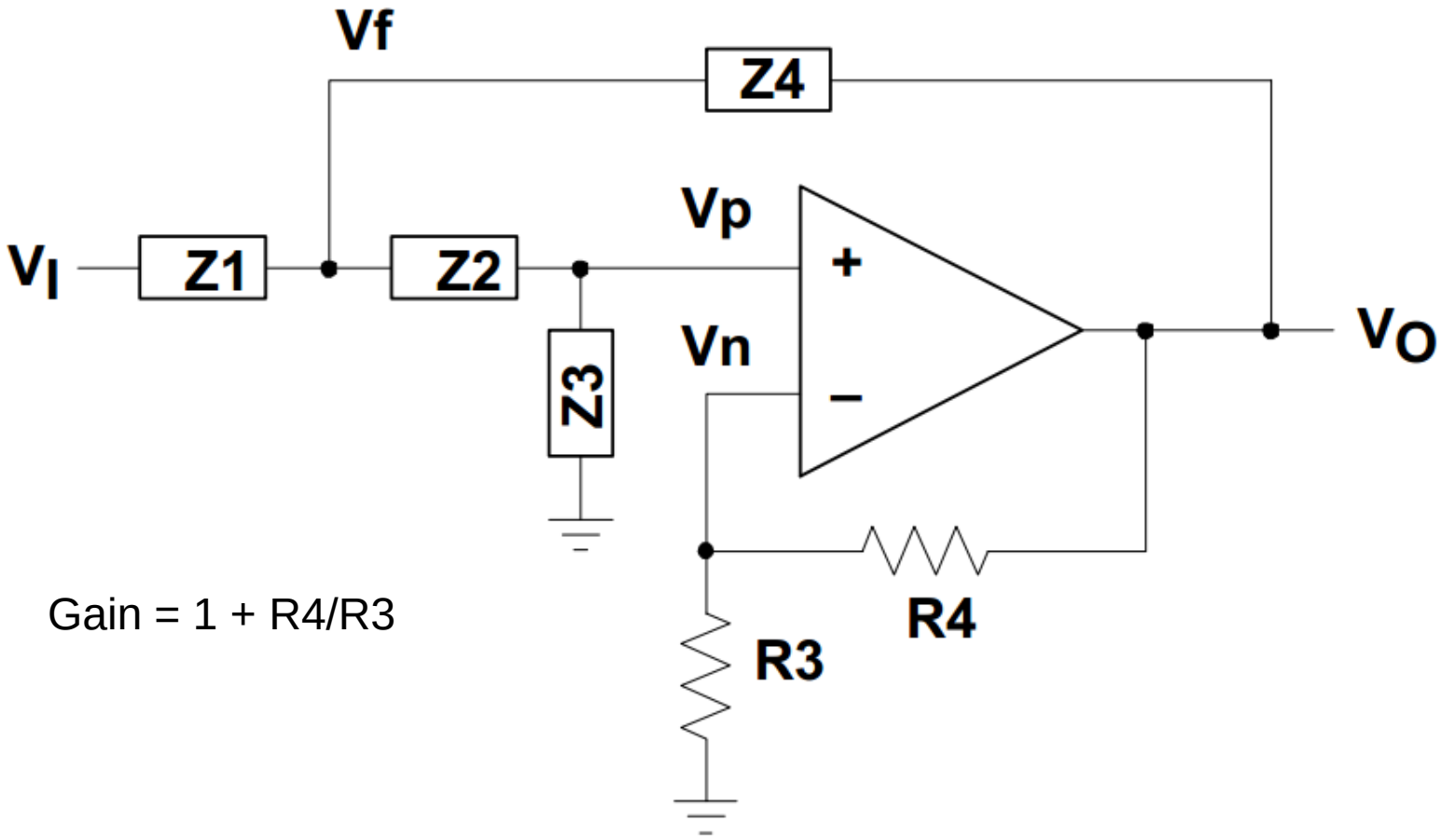


Filter Stage: 1

Passband Gain(Ao): 3



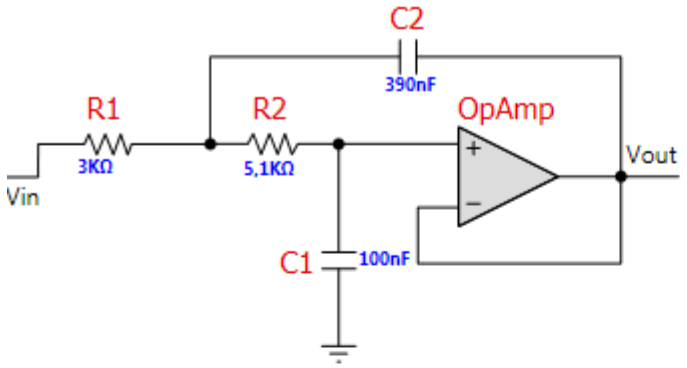
Sallen-Key



Gain = 1 + R4/R3

Demonstratie: http://www.ti.com/lit/an/sbfa001c/sbfa001c.pdf
http://www.ti.com/lit/an/sloa024b/sloa024b.pdf

FTJ Activ de ord 2, gain 1, 200Hz (Sallen-Key)

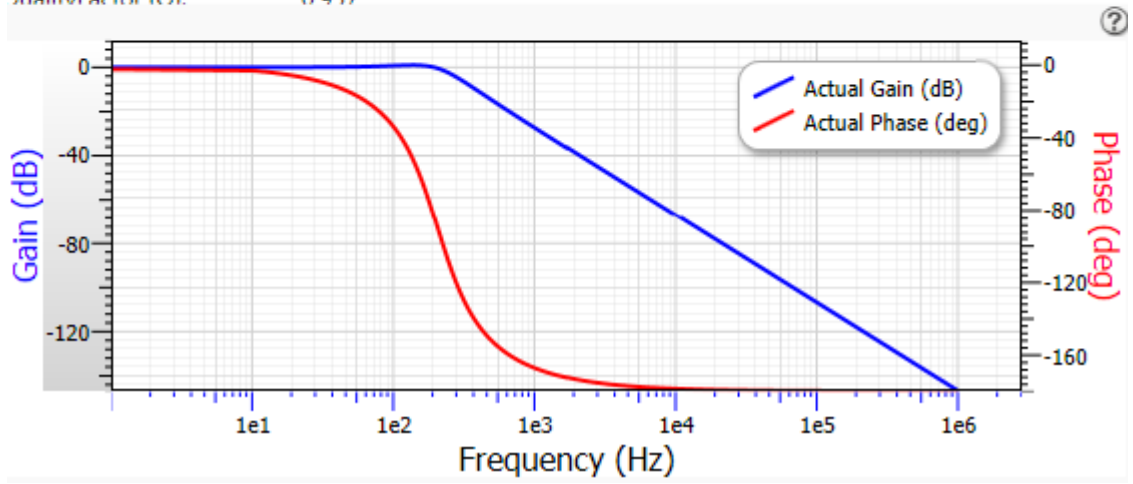


Filter Stage: 1

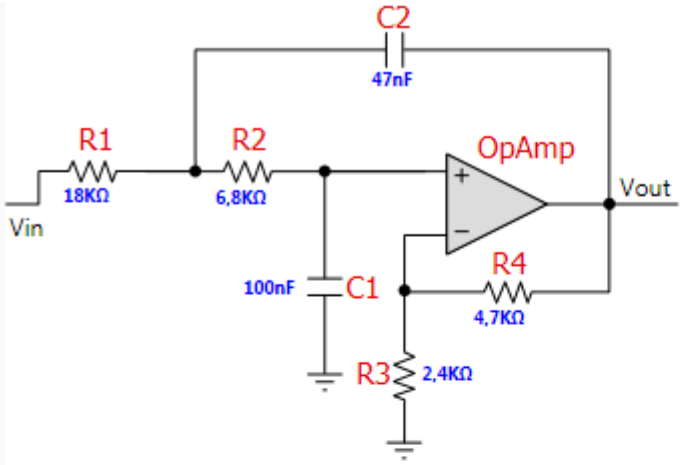
Passband Gain(Ao): 1

Cutoff Frequency(fn): 210,001 Hz

QualityFactor (Q): 0.957



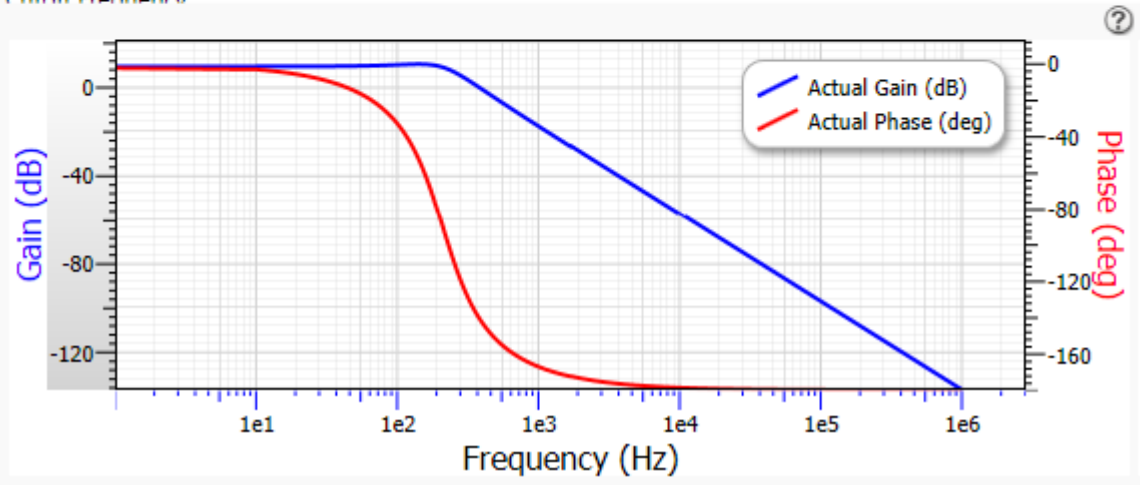
FTJ Activ de ord 2, gain 3, 200Hz (Sallen-Key)



Filter Stage: 1

Passband Gain(Ao) : 3

Cutoff Frequency



Filtru trece sus

Filtru trece sus

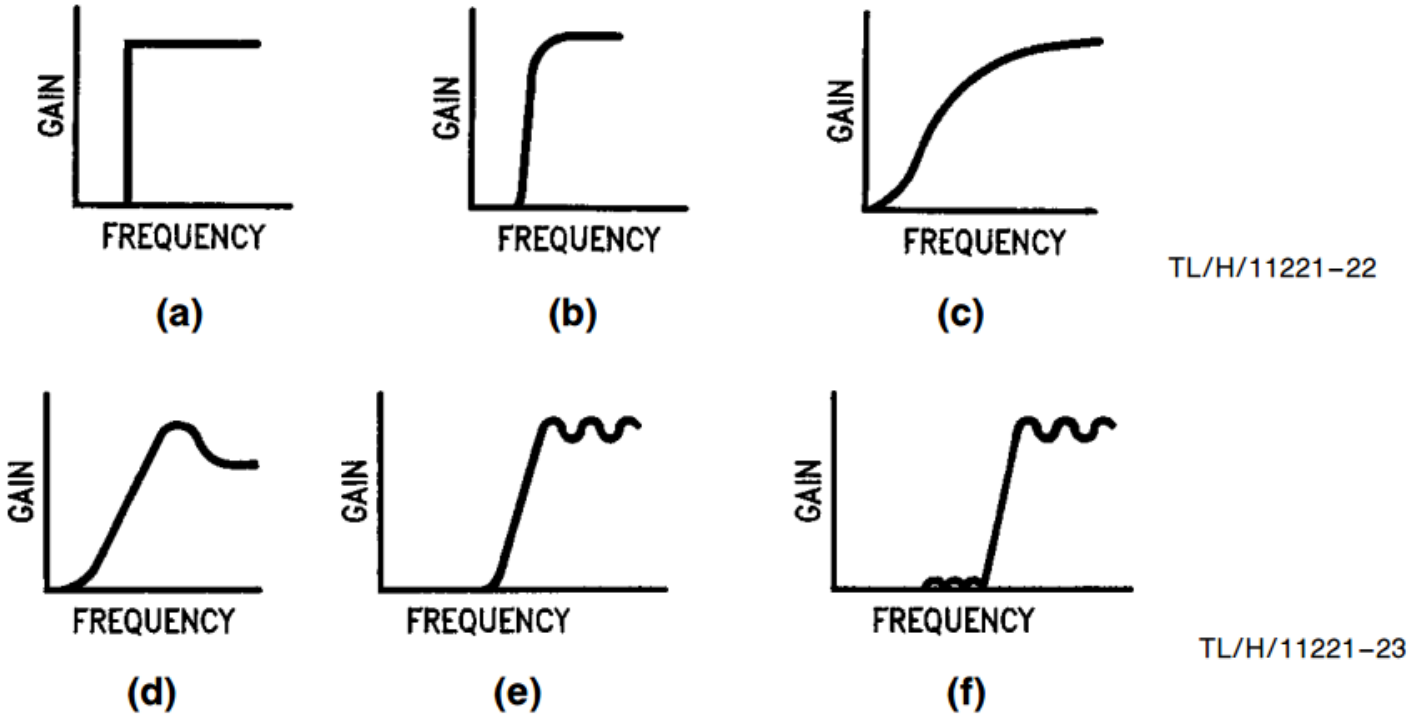
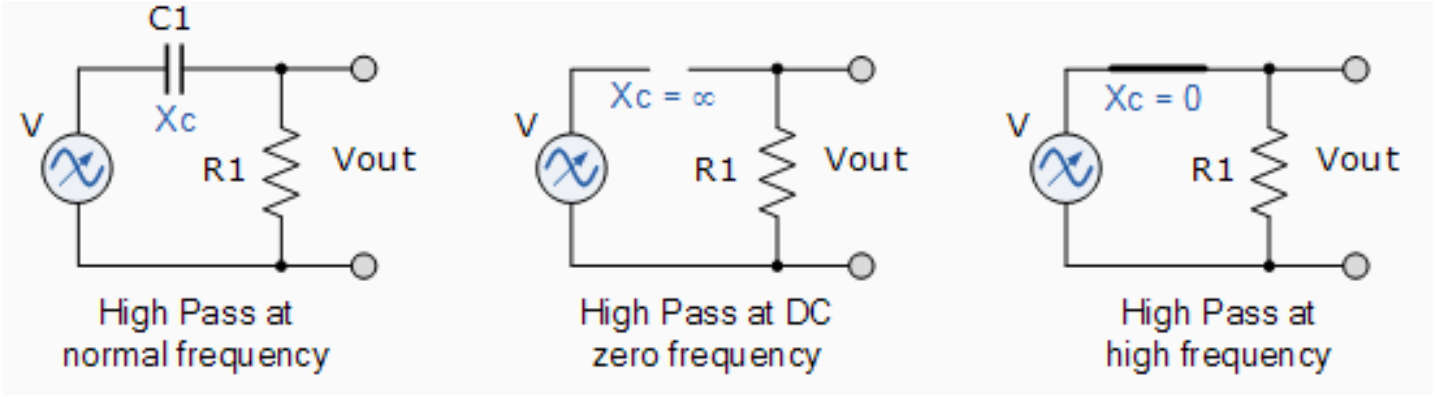


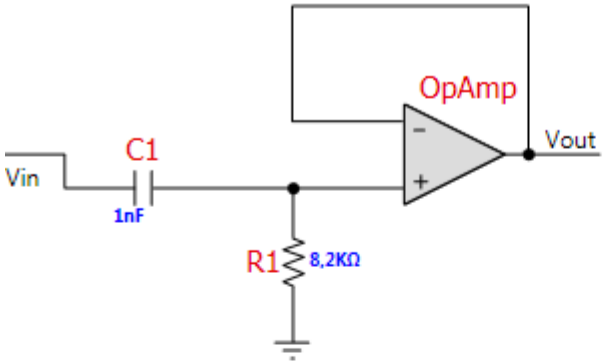
FIGURE 14. Examples of High-Pass Filter Amplitude Response Curves

FTS (HPF) pasiv de ordin 1



$$V_{out} = V_{in} \frac{R}{\sqrt{R^2 + {X_c}^2}} unde X_c = \frac{1}{2\pi f C}$$

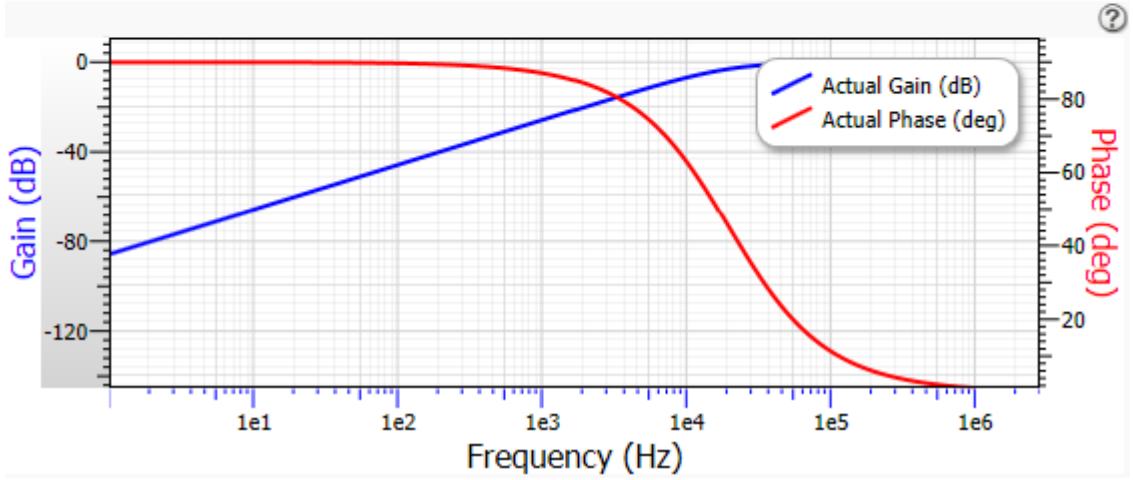
FTS ord 1, cu buffer, 20kHz



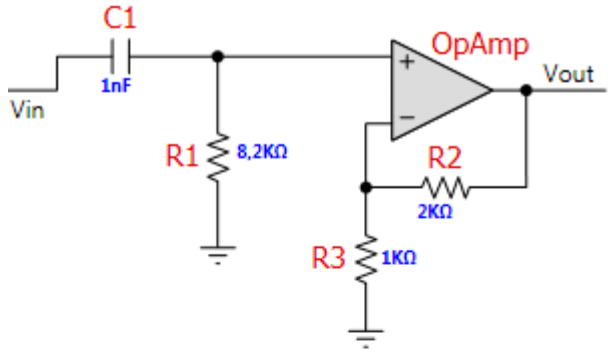
Filter Stage: 1

Passband Gain(Ao) : 1

Cutoff Frequency (fn): 20 kHz

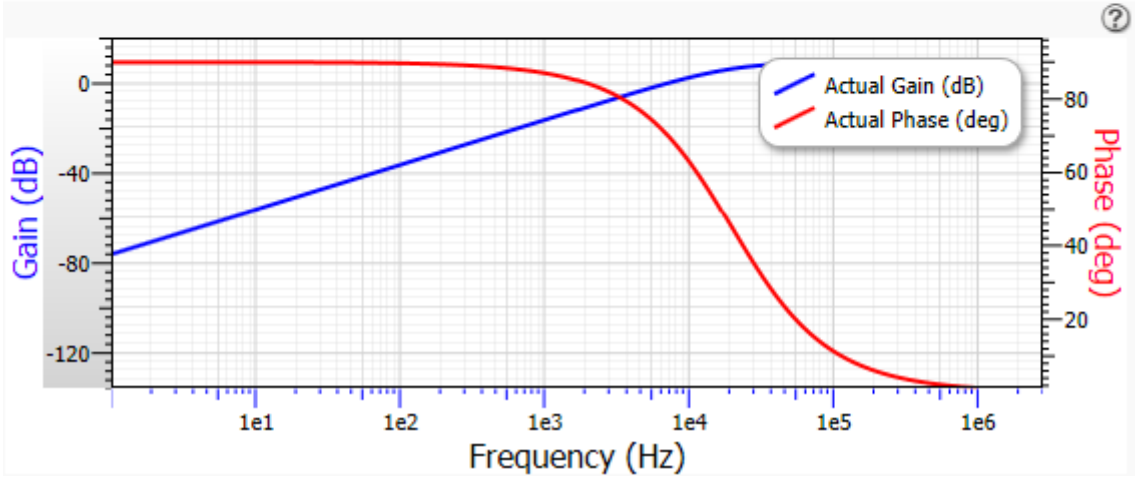


FTS ord 1, gain 3, 20kHz

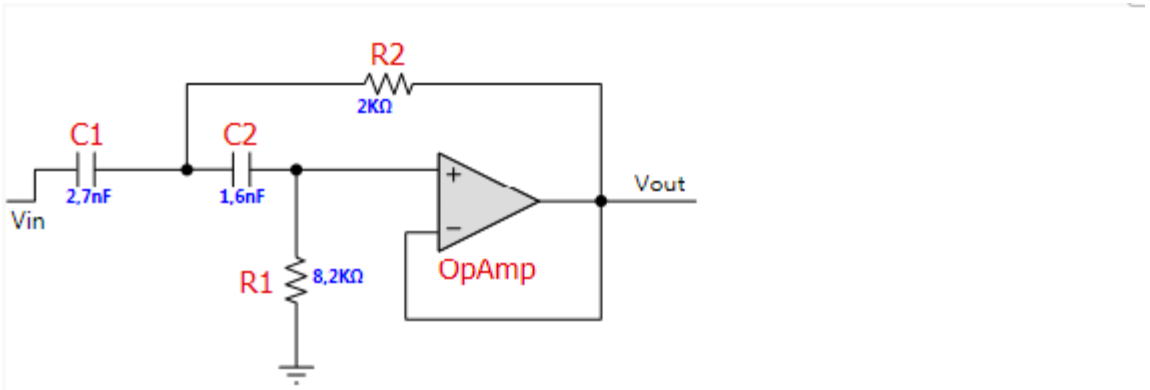


Filter Stage: 1

Passband Gain(Ao) : 3



FTS Activ de ord 2, gain 1, 20kHz (Sallen-Key)

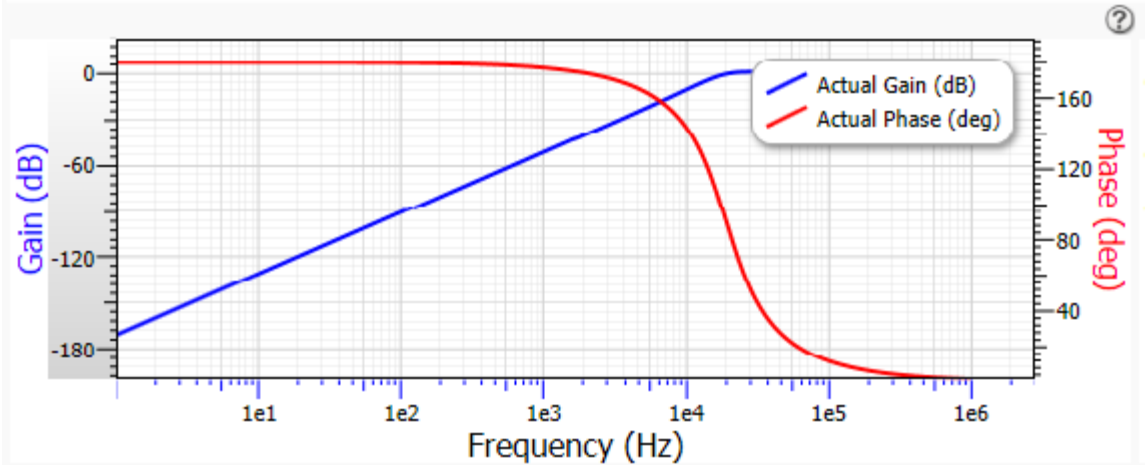


Filter Stage: 1

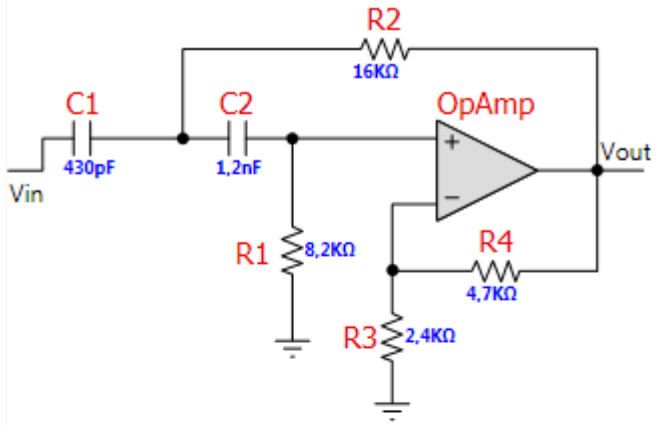
Passband Gain(Ao) : 1

Cutoff Frequency(fn): 19,0475 kHz

QualityFactor (Q): 0,957



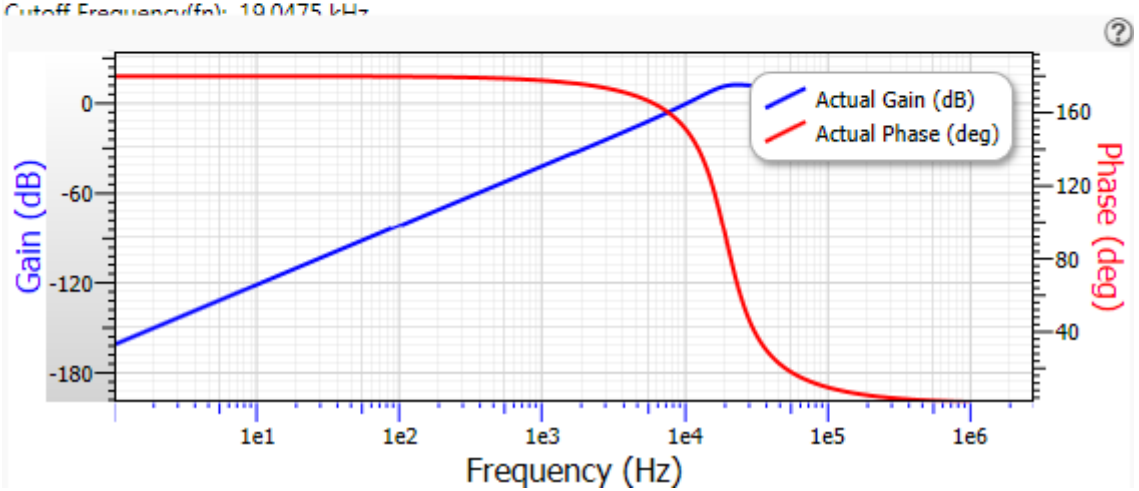
FTS Activ de ord 2, gain 3, 20kHz (Sallen-Key)



Filter Stage: 1

Passband Gain(Ao) : 3

Cutoff Frequency(fp): 19.0475 kHz



Filtru trece bandă

Filtru trece bandă

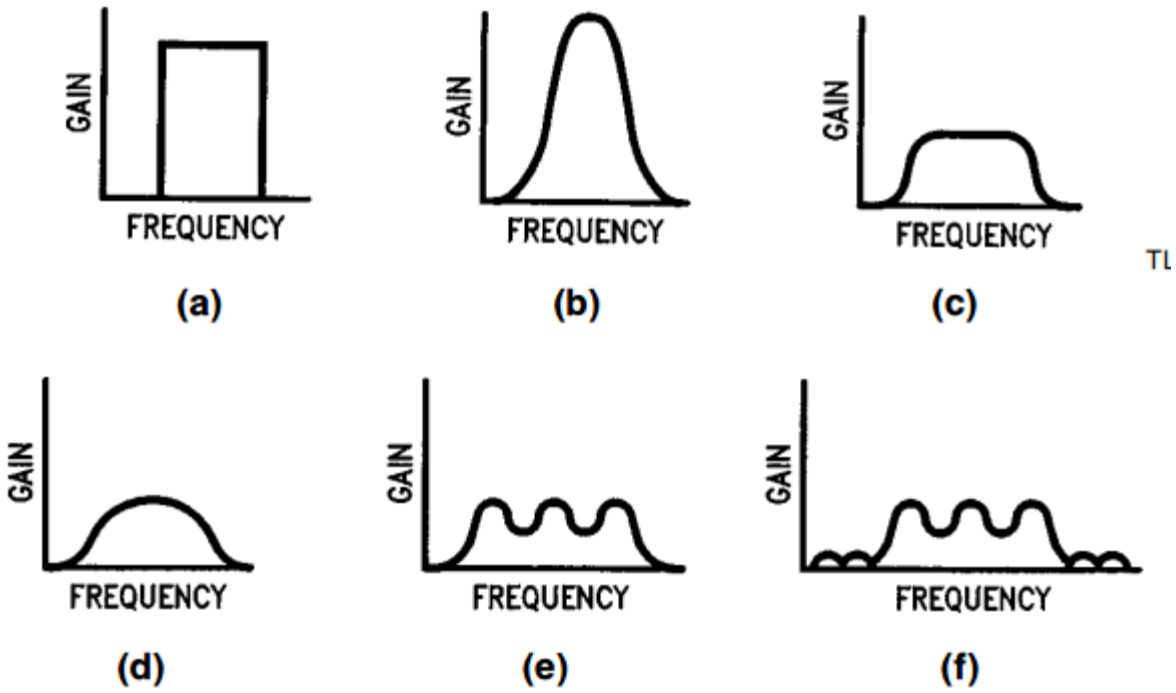
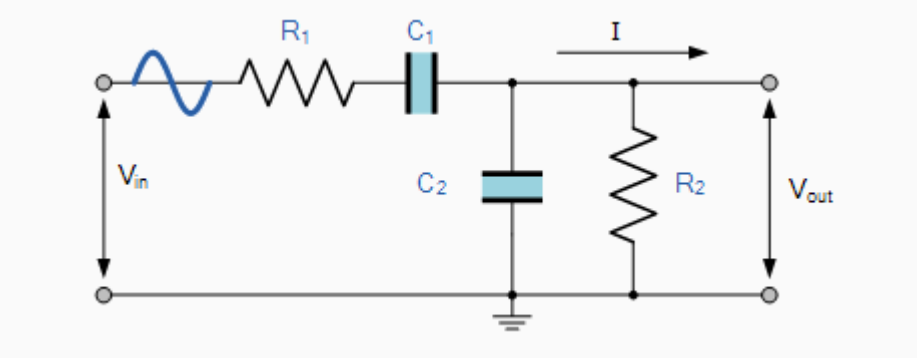
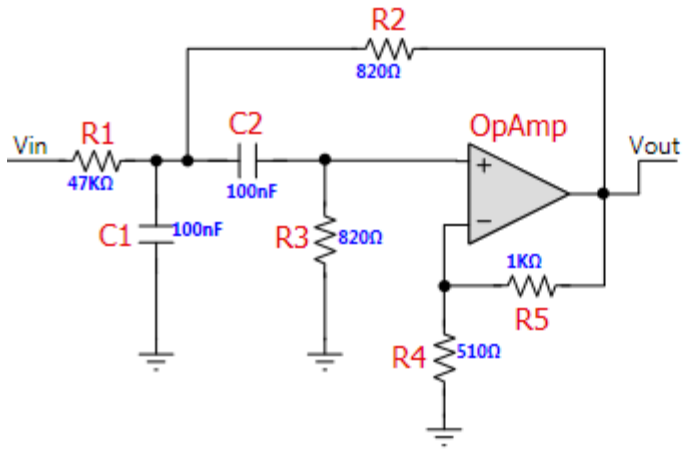


FIGURE 5. Examples of Bandpass Filter Amplitude Response

FTB (BPF) pasiv de ordin 1

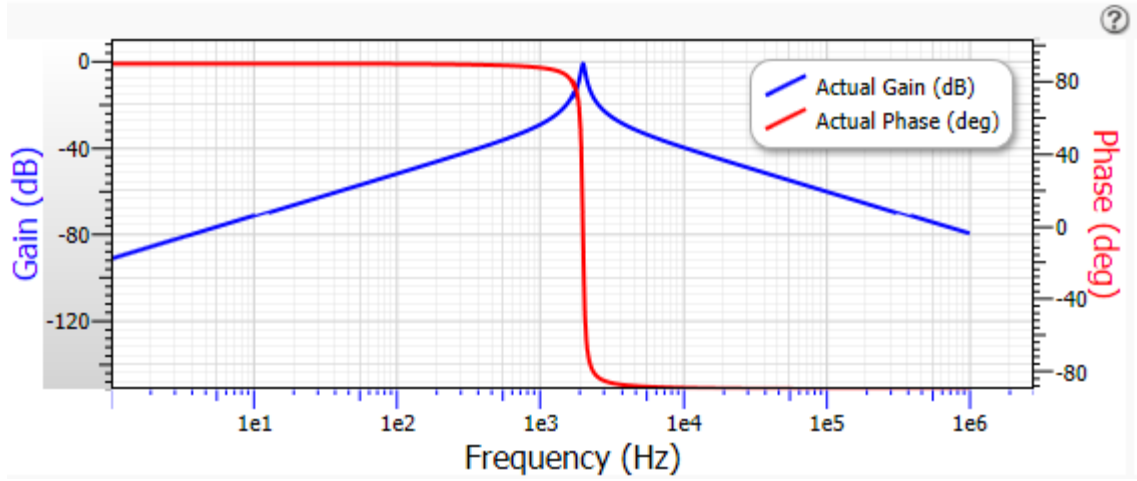


FTB Activ de ord 2, gain 1, 2kHz cu 100Hz

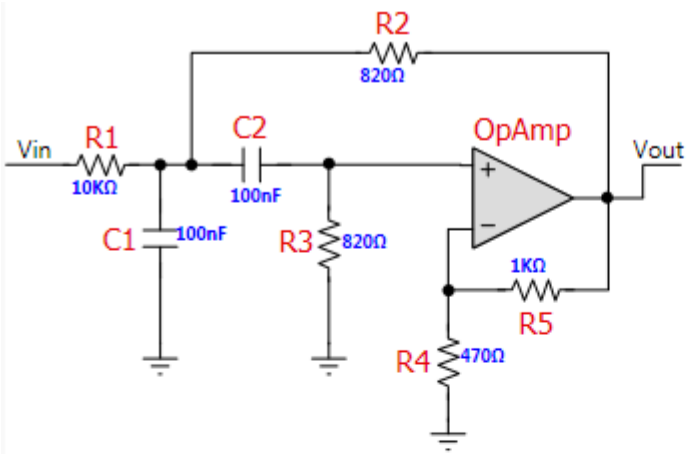


Filter Stage: 1

Passband Gain(Ao) : 1

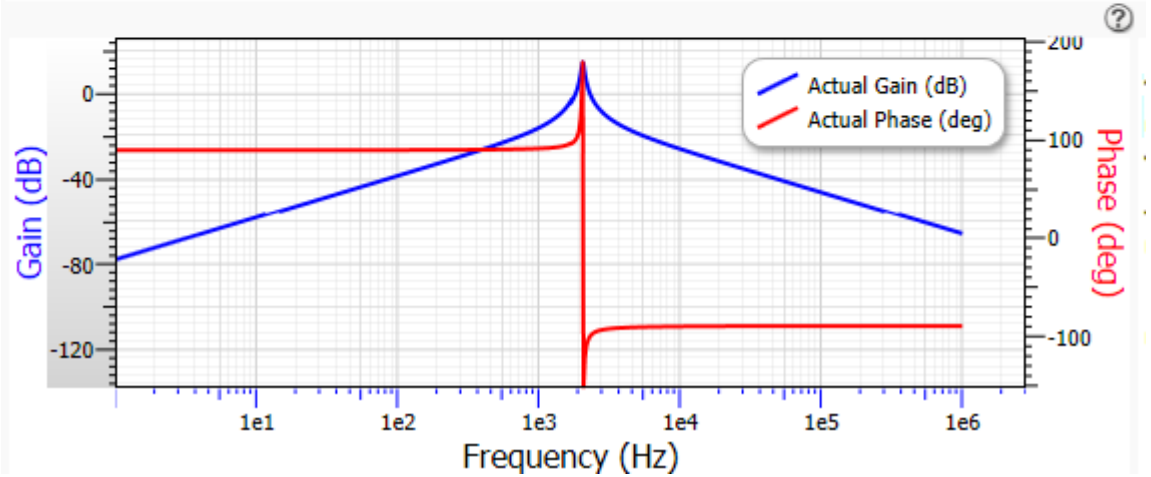


FTB Activ de ord 2, gain 5, 20kHz cu 100Hz

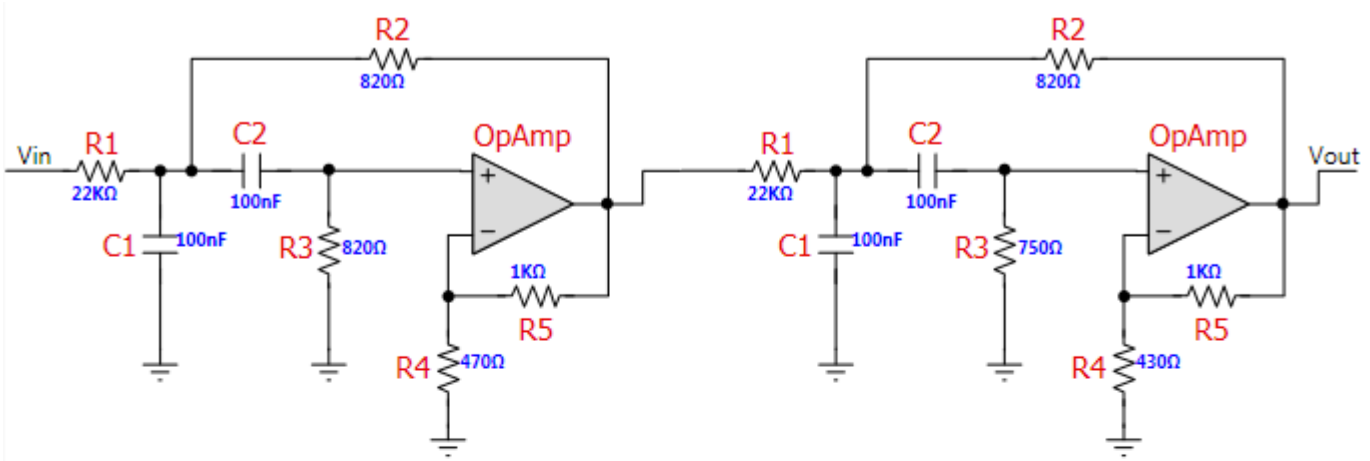


Filter Stage: 1

Passband Gain(Ao) : 5



FTB Activ de ord 4, gain 5, 20kHz cu 100Hz

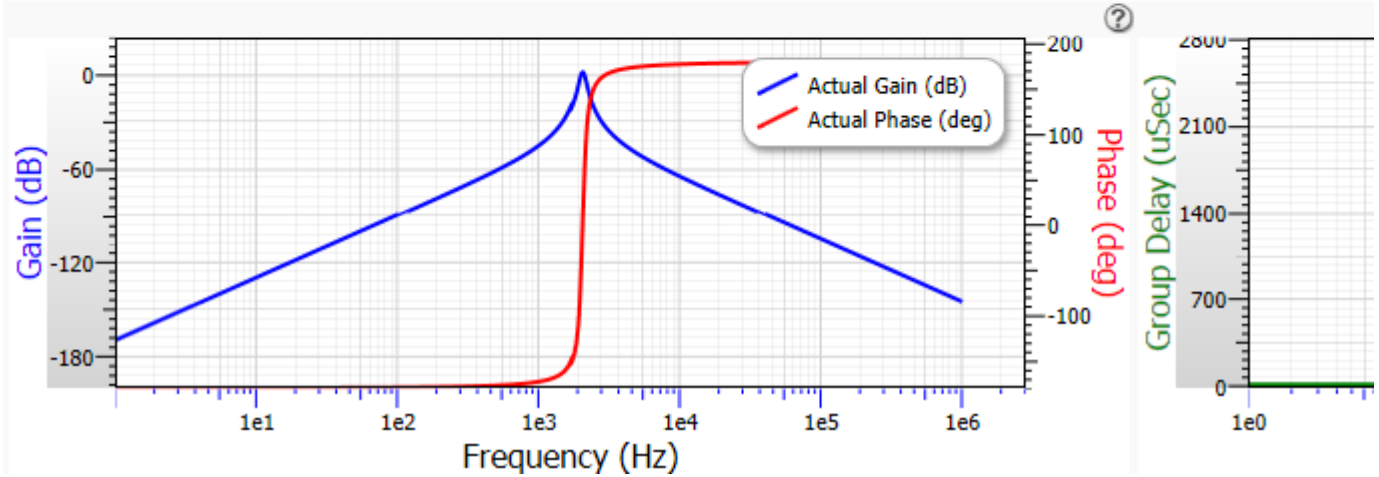


Filter Stage: 1

Passband Gain(Ao) : 2,236

Filter Stage: 2

Passband Gain(Ao) : 2,236



Filtru rejectează bandă

Filtru rejectează bandă

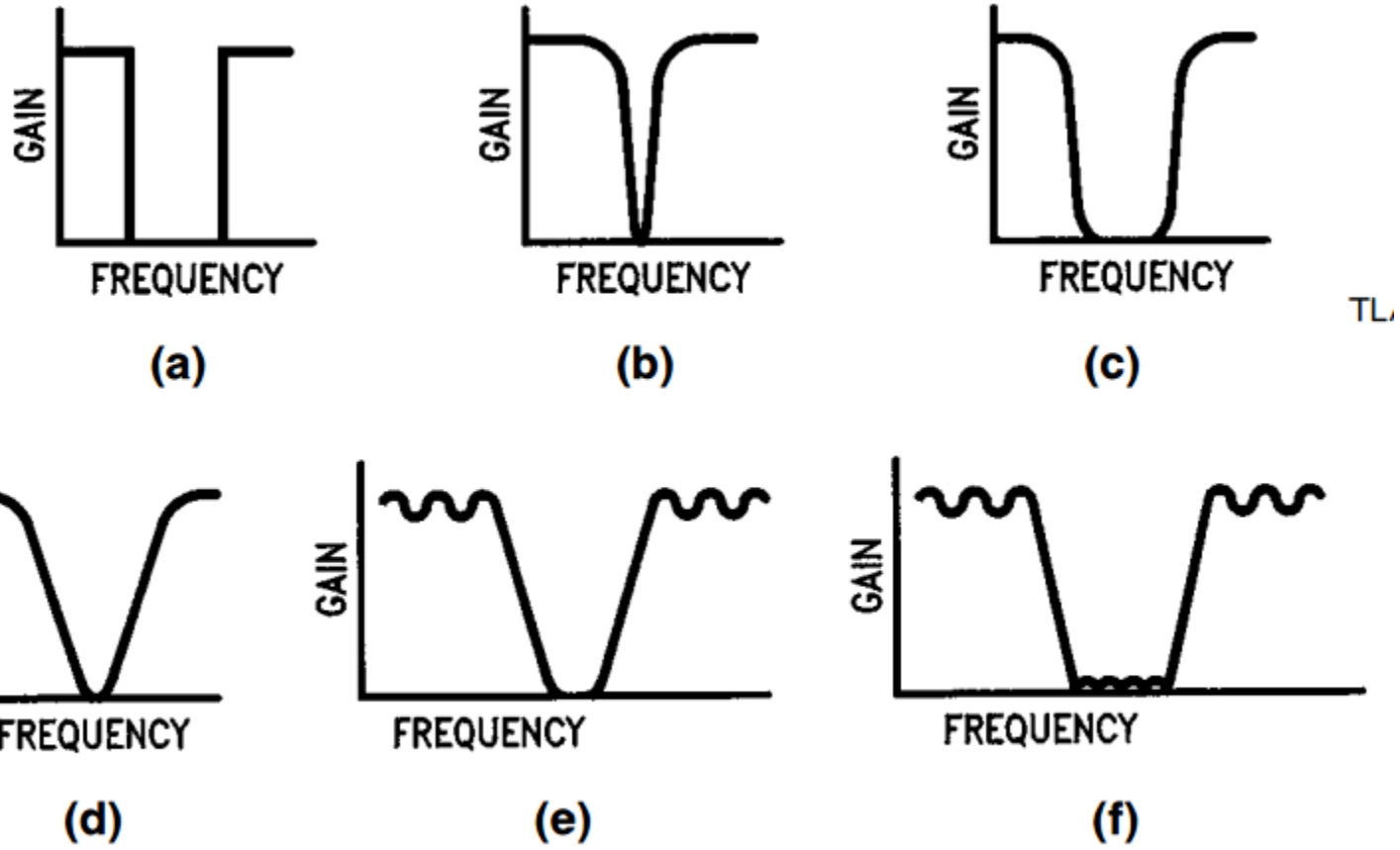
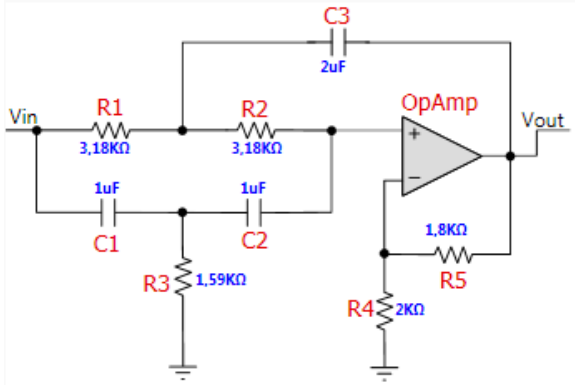
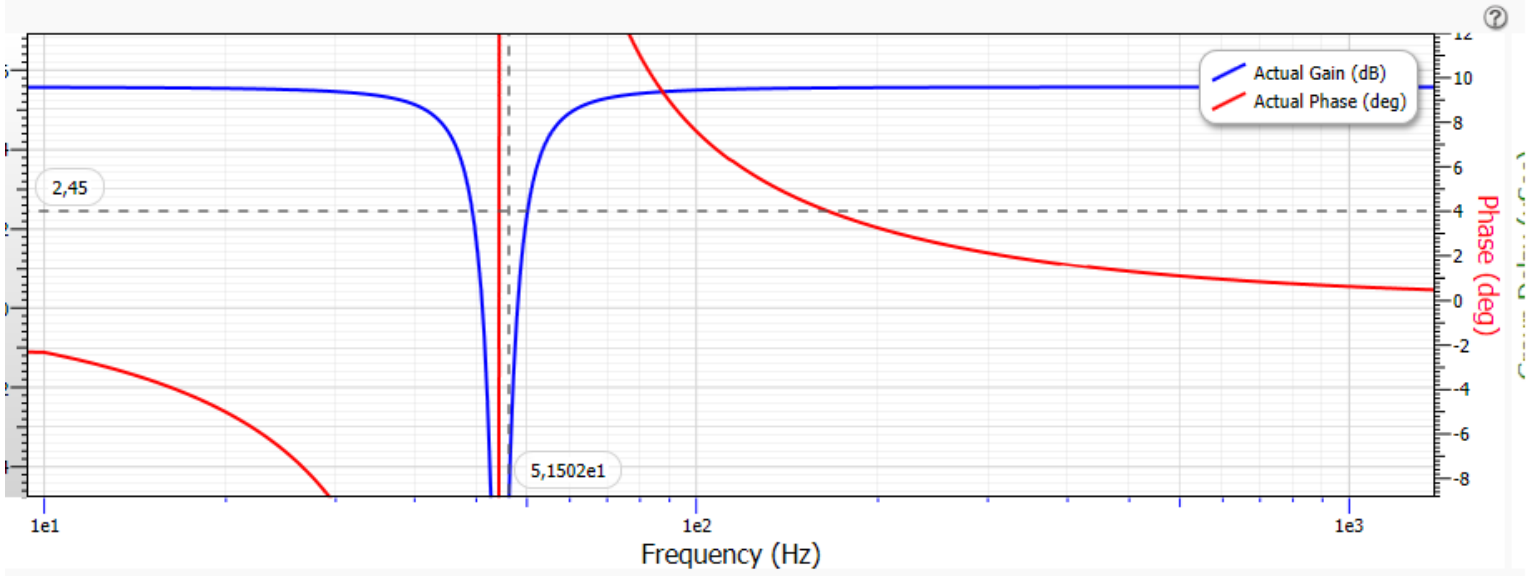


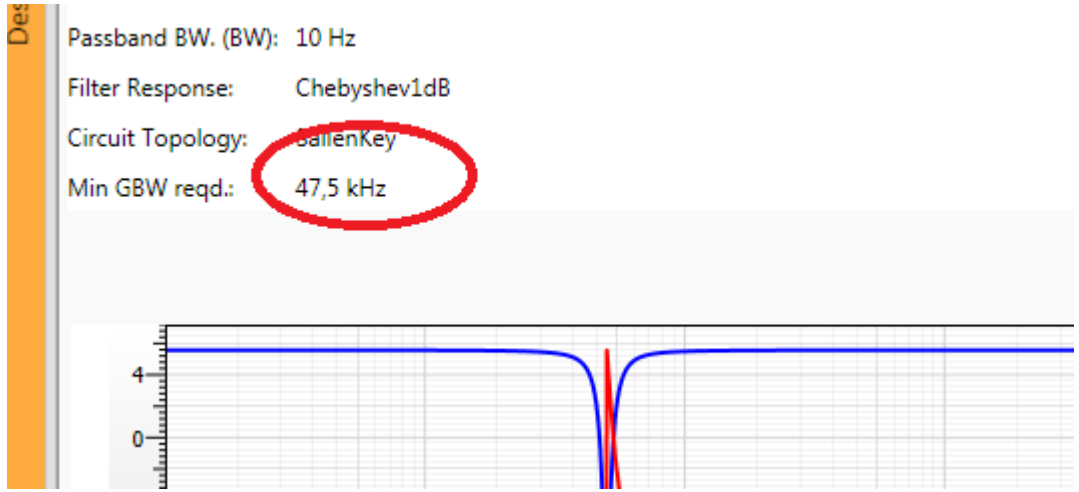
FIGURE 8. Examples of Notch Filter Amplitude Responses

FRB Activ de ord 2, gain 1, 50Hz cu 10Hz





Observații legate de AO-ul ales



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